

Risk Perception: Model, Mechanisms, and Market Outcomes*

Xindi He

Yucheng Liang

Georgia Institute of Technology

Carnegie Mellon University

October 4, 2026

Abstract

Prior research identifies many price-path features that shape investor risk perception, but their correlation and conceptual overlap complicate interpretation and raise concerns about overfitting and generalizability. Using a data- and literature-driven approach, we develop a parsimonious model based on volatility, total return, volatility recency, price range, and return autocorrelation. The model generalizes across samples and risk perception measures. Diagnostic experiments examine the channels and mechanisms behind these features' effects. Applying the experimental weights to market data, we find that the resulting index is associated with option-implied volatility, subsequent realized volatility, retail net selling, and large mutual-fund redemptions.

JEL classification: C91, G11, G12, G41.

Keywords: risk perception, price charts, feature selection, lab experiments, beliefs.

*He: Scheller College of Business, Georgia Institute of Technology, xindi.he@scheller.gatech.edu. Liang: Tepper School of Business, Carnegie Mellon University, ycliang@cmu.edu. We thank Alex Imas and Stefan Zeisberger, seminar participants at Emory University, the PBC School of Finance at Tsinghua University, and the University of Chicago Booth School of Business, and conference participants at FIRS 2025 for helpful comments. We thank Yiyang Zhang for excellent research assistance. All errors are our own. All experiments were reviewed by the Institutional Review Board of the Georgia Institute of Technology and determined to be exempt under 45 CFR 46.104(d)(2), protocol H23524 ("Risk Perception"), approved on January 4, 2024. Every experiment was pre-registered on AsPredicted before data collection; the registration for each is cited where that experiment is first described.

I Introduction

Risk is a central concept in finance: it shapes investment decisions and is a fundamental determinant of asset prices. Standard theory typically takes the variance of an asset’s return as the measure of its risk, following Markowitz (1952), and does not separately model how investors perceive this risk; empirical work proxies for it with realized past volatility.¹ A growing literature instead measures risk perception directly and finds that it departs systematically from realized volatility, that it matters for investment decisions, and that a wide array of price-path features shape it (Duxbury and Summers, 2018; Holzmeister et al., 2020; Borsboom and Zeisberger, 2020; Zeisberger, 2024; Bose et al., 2022).

However, many of these features are highly correlated and conceptually overlapping, creating three challenges for modeling risk perception. First, a model with many features may not generalize well across settings because it is prone to overfitting. Second, high correlation between features may destabilize the estimated coefficients, obscuring the effect of each on perceived risk. Third, when features overlap conceptually, it is hard to pinpoint the channels and mechanisms underlying any given effect. A separate question is whether a model estimated from experimental risk judgments is informative about market outcomes. We address the three modeling challenges with a series of experiments and then apply the resulting model to market data.

To construct a parsimonious model of risk perception that generalizes well, we first run an online experiment that measures participants’ risk perception after viewing anonymized stock price charts. Participants are recruited on Prolific, an online platform widely used for academic surveys, and screened for investment experience. In each round, the participant sees a chart of a stock’s month-end price over a twelve-month window, with the final price normalized to 100. Some rounds show real U.S. stocks; the rest are synthetic, built by reshuffling a fixed sequence of twelve monthly returns, following Nolte and Schneider (2018), and, in some cases, shifting every return by a

¹Risk is sometimes also defined by an asset’s return covariance with systematic factors, which determines the non-diversifiable part of its return variance. Nevertheless, Chinco et al. (2022) find in survey experiments that investors’ allocations respond to an asset’s mean and volatility but not to its correlation with consumption growth, the risk factor that consumption-based models emphasize.

constant. Synthetic charts vary the ordering of the returns while holding volatility and every higher moment of the return distribution fixed by construction.

Participants answer four questions about each price chart. The *risk score* asks directly how risky the stock is, on a 0–10 scale. This risk score is not incentivized, but it allows participants to report an overall assessment without restricting which aspects of risk they consider. To complement the risk score, we also elicit two belief measures about the future return distribution: participants give a price *forecast* for the three-month horizon and report their confidence in that forecast, measured as the probability that the realized price falls within five points of their forecast. We define *forecast uncertainty* as 100 minus this reported probability. Both elicitations are incentivized for accuracy. Finally, we elicit an *investment* decision, measured as the share of a \$3 endowment that the participant allocates to the stock. This incentivized choice allows us to assess how strongly the risk score and elicited beliefs are reflected in participants’ investment decisions. The bonus is determined by a single randomly selected incentivized question about a real chart (Azrieli et al., 2018). Since participants are not told which charts are real, it is optimal for them to treat every chart as real.

We select features using the risk score, the comprehensive but unincentivized measure, for two empirical reasons. First, the risk score predicts the incentivized investment decision better than the forecast and forecast uncertainty combined. Second, participants’ risk scores vary much more systematically across charts than their forecast uncertainty does, making the risk score more informative for distinguishing among competing feature models. We then use the incentivized belief measures to evaluate whether the selected model carries across measures and to study the channels through which the selected features affect perceived risk.

We design the selection procedure to be driven by the literature and the experimental data. The candidate set includes 23 features collected from the literature and adapted to our setting. As many features are highly correlated, we first group them by agglomerative clustering on their chart-level correlations, which yields eleven clusters, and we keep the most representative member of each.²

²We choose the representative to be the medoid, which is the member feature with the highest average absolute correlation with the other members of the same cluster, with one exception. In the cluster containing total return, the

We then search exhaustively over all linear models that can be built from the eleven representatives, requiring only that volatility—the standard measure of risk—be included, and rank them by cross-validated R^2 on held-out charts. This ranking criterion penalizes overfitting and lets us select models that are more generalizable. The best model contains six features: volatility, total return, volatility recency, price range, return autocorrelation, and the number of months the price stays below its starting price. Since the last feature is correlated at -0.80 with total return, and its coefficient flips sign depending on whether total return is controlled for, we drop it and keep the remaining five features. This five-feature model has a cross-validated R^2 of 0.187 , almost 10 times that of volatility alone (0.019), and ranks second among all volatility-including five-feature models.

The five-feature model also performs well on forecast uncertainty, a narrower but incentivized measure of perceived risk elicited in the same experiment. On this outcome, it ranks first among all 1,024 volatility-including models constructed from the eleven cluster representatives, with a cross-validated R^2 of 0.013 . It also outperforms both the six-feature specification that performs best on the risk score and the full 23-feature model.

The parsimony and conceptual independence of the five features allow us to ask *why* each one matters. Section III addresses this question with six additional experiments on independent samples. We begin with the *Uncertainty* and *Downside* experiments, which isolate two specific belief channels. The former elicits forecast uncertainty, while the latter elicits the perceived probability of a large downside forecast error. Both experiments also use designed synthetic charts that vary individual features as cleanly as possible, allowing their effects to be examined through pairwise comparisons. Volatility recency, price range, and autocorrelation affect both belief measures, and volatility affects forecast uncertainty. Total return stands apart. It has the largest effect on the risk score in the main experiment (-0.230), yet its effect is essentially zero in both the *Uncertainty* and *Downside* experiments.

One potential explanation is that a history of falling prices makes participants more concerned about losses on a long position, and that this concern is reflected in the risk score. We test this

medoid is the average price, but it depends on where the price path sits inside the plot window. As a result, we replace it with total return, which is invariant to that placement.

explanation with the *Position experiment*, in which we randomly assign participants long or short positions before they rate the charts. A future price decline hurts a long position but benefits a short position. Yet past declines raise risk scores in both groups, suggesting that the total-return effect is not simply a response to losses on the participant's own position.

The volatility-recency effect admits two explanations: participants may deliberately view the recent past as more informative about future risk, or recent returns may receive greater weight simply because they are viewed first. The *Chosen-order* and *Forced-order* experiments distinguish between these explanations. In the *Chosen-order experiment*, the chart starts concealed and participants reveal the twelve monthly returns in an order of their choice, under a randomly assigned cap that gives them an incentive to reveal the returns they consider most informative first. Reveal order strongly tracks recency: in 34% of rounds with at least three reveals, the first three are the three most recent returns in recency order, while return direction and turning points matter much less. This pattern suggests that the weight on recent volatility is at least partly deliberate. In the *Forced-order experiment*, the computer instead reveals the twelve returns in random order, breaking the link between calendar recency and viewing order. The effect of calendar recency remains essentially unchanged, while reveal order itself has no detectable effect on forecast uncertainty. Thus, the greater weight on recent volatility does not appear to arise simply because recent returns are viewed first.

The last two features, price range and return autocorrelation, are empirically correlated but conceptually distinct. Price range is a property of the price path, whereas return autocorrelation is a property of the return sequence. To distinguish their mechanisms, the *Return-arrow experiment* replaces the price line with the twelve monthly returns, each drawn as an arrow from a common zero line, so that no price level is displayed. Removing the price line largely eliminates the range effect while leaving the autocorrelation effect essentially unchanged. This contrast indicates that the range effect depends on the visual price path, whereas the autocorrelation effect reflects the ordering of returns and survives when price levels are removed.

The final part of the paper examines whether the experimental model is informative about market

outcomes. We take the five-feature model estimated in the main experiment to real market data. For each asset, we compute the five features from its trailing monthly returns and combine them using the coefficients estimated from experimental risk scores. This yields the *chart-risk index* S^E , the risk score that the experimental model would assign to the asset's recent price path. We also construct the *sequence index* S^S from volatility recency, price range, and return autocorrelation, the three features that can change when the same returns are reordered. Both indices use only the experimental weights; no market outcome enters their construction. We then ask whether the indices behave like market measures of risk and whether market investors respond to them as participants in our experiments did.

We first examine option-implied volatility in U.S. stocks. Each of the five features is associated with implied volatility in the direction predicted by the experiment. A one-standard-deviation increase in the chart-risk index is associated with 0.15 standard deviations higher contemporaneous log implied volatility, and the sequence index is also positively associated with implied volatility after controlling for volatility and total return. Both indices predict next-month implied volatility conditional on lagged implied volatility and realized volatility over the following year conditional on current implied volatility.

We next ask whether market demand exhibits the same pattern as investment choices in the experiment: lower demand for stocks perceived as riskier. The chart-risk index predicts retail net selling over the following year. We also apply the index to mutual-fund share classes using their trailing three-year net asset value (NAV) paths. A one-standard-deviation increase in the index is associated with a 4.7-percentage-point higher probability of a large redemption—a net outflow of more than 20 percent of assets—over the following year. The sequence index also predicts retail selling where retail investors account for a larger share of trading and predicts mutual-fund redemptions after controlling for volatility and total return.

Taken together, these results indicate that a model of risk perception estimated in the laboratory is informative about how risk is priced and traded outside it.

Our work is closest to the experimental literature on risk perception from price paths. This

literature shows that perceived risk responds not only to the distribution of returns, but also to their ordering, salient features of the resulting price path, and how that path is displayed (Holzmeister et al., 2020; Zeisberger, 2024; Borsboom and Zeisberger, 2020; Nolte and Schneider, 2018; Duxbury and Summers, 2018; Bose et al., 2022; Weber et al., 2005; Kaufmann et al., 2013; Diacon and Hasseldine, 2007; Unser, 2000). We contribute on three fronts.

First, we develop a parsimonious model of risk perception that is selected rather than assumed. The selection is data-driven, in the spirit of recent work that identifies the most informative predictors from large candidate sets (Enke and Shubatt, 2023; Guan et al., 2025). The resulting five-feature model generalizes beyond the sample and measure used for selection, providing a compact mapping from price paths to perceived risk that can be taken to other settings.

Second, we examine the channels and psychological mechanisms underlying the effects of the five features. Prior work shows that visual salience in price charts predicts perceived risk (Bose et al., 2022). We extend this evidence with a series of targeted experiments that separately elicit beliefs and randomly vary participants' investment positions, the order in which returns are revealed, and the display format of price histories. These experiments allow us to test whether the feature effects operate through specific beliefs, distinguish competing explanations for the total-return and volatility-recency effects, and show that price range and autocorrelation arise from different mechanisms. More broadly, our approach connects to recent work on the cognitive mechanisms behind risky choice and valuation (Khaw et al., 2021; Enke and Graeber, 2023; Oprea, 2024; Bohren et al., 2025; Shubatt and Yang, 2026a,b; Dean et al., 2026) and belief formation (Hartzmark et al., 2021; Ba et al., 2026; Augenblick et al., 2025; Jiang et al., 2025; Bordalo et al., 2026; Fan et al., 2026).

Third, we take the experimentally estimated model of perceived risk to market data. The five-feature model is associated with option-implied volatility, subsequent realized volatility, retail selling, and mutual-fund redemptions. Prior work shows that price paths affect investor satisfaction and selling (Grosshans and Zeisberger, 2018; Bansal and Jacob, 2022), that volatility adaptation shapes perceived risk and asset prices (Payzan-LeNestour et al., 2023), and that a chart-shape

score validated in the laboratory predicts returns in the cross-section (Cordes et al., 2021). Our contribution is to estimate a joint model of risk perception from experimental judgments, hold its weights fixed, and apply it across several distinct market settings. This approach also differs from Jiang et al. (2023), who train image models directly on price charts to predict returns: we begin with experimentally measured risk perceptions and ask whether the resulting model is informative about option-implied volatility, subsequent realized volatility, and investor flows.

II Measuring and Modeling Risk Perception

II.A The main experiment

Price charts. The main experiment measures how a stock’s past price path shapes perceived risk and provides the data used to construct our model. We run the experiment on Prolific, where participants view stock price charts and report the stock’s perceived risk, their beliefs about its future price, and how much of an endowment they would invest in it. Each participant completes twenty rounds. In each round, the participant views a stock’s month-end price over a twelve-month window, with the final price normalized to 100. Figure 1 shows an example.³

Ten of the twenty charts show real U.S. stocks, and ten are *synthetic*. Participants are not told which charts are real, but they know that only responses to real charts determine their bonus. It is therefore optimal for them to treat every chart as real.

The ten real charts are drawn at random from a pool of 100 charts constructed from CRSP monthly data over February 2019–January 2020. To form the pool, we randomly select stocks with a mean monthly return between -1.35% and $+1.35\%$ and a monthly return standard deviation between 4.05% and 6.07% over this period.⁴ We also require price data through April 2020 so that the three-month-ahead realization is observed for bonus payments.

³We normalize the final price for two reasons. First, it makes a given expectation of future prices correspond to the same expectation of future returns in every round. Second, because the plot window is centered vertically on the final price of 100, the visible price range is identical across rounds, eliminating variation in the plotting window as a source of framing.

⁴The selected price paths lie within the plot window, which runs from 70 to 130.

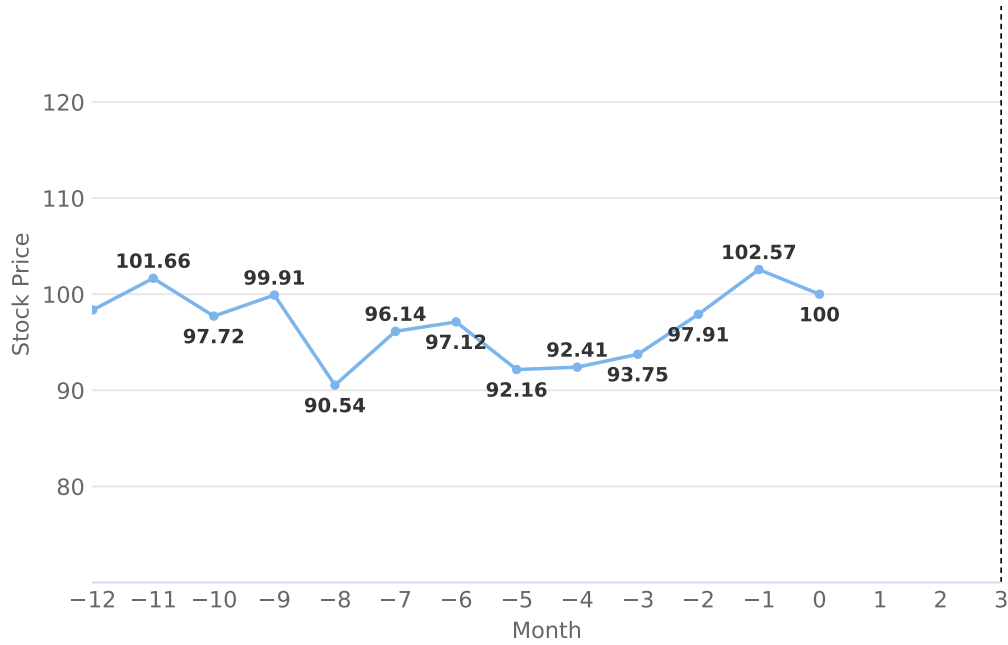


Figure 1. Example price chart in the main experiment.

This figure shows an example chart from the main experiment. It plots thirteen month-end prices from Month -12 to Month 0 , normalized so that the final price is 100. All prices after the initial observation are labeled. The price axis ranges from 70 to 130, and the dashed line at Month 3 marks the forecast horizon.

Alt text: Line chart of stock prices from Month -12 to Month 0 , fluctuating between roughly 91 and 103 and ending at 100. A dashed line at Month 3 marks the forecast horizon.

The ten synthetic charts are drawn at random from a second pool of 100 charts constructed by reshuffling one of three fixed *base sequences* of monthly returns, following Nolte and Schneider (2018). Base sequence 1 consists of the twelve monthly returns -10 , -5 , -2.5 , -2.5 , -1.25 , -1.25 , $+1.25$, $+1.25$, $+2.5$, $+2.5$, $+5$, and $+10$ percent, which have a mean of zero and a standard deviation of 5.06%. The other two base sequences are formed by shifting every return in base sequence 1 by $+1.125$ and -1.125 percentage points, respectively. These shifts change the mean return but leave volatility, skewness, and kurtosis unchanged. Within each base sequence, reshuffling changes the ordering of the same twelve returns and the resulting path features while holding the return distribution fixed.

Measuring risk perception. Risk perception is inherently subjective and multifaceted. A direct risk rating can capture this broad judgment, but it has no objectively correct value and therefore

cannot be incentivized. Beliefs about specific aspects of the future return distribution can instead be incentivized, but each captures only one potential component of perceived risk. We therefore use both: an unincentivized risk score as a comprehensive measure and incentivized belief measures to isolate specific channels.

Specifically, for each chart we first elicit the comprehensive measure—the *risk score*—by asking participants to answer the question “How risky do you think this stock is?” on a 0–10 Likert scale. We then ask participants to forecast the stock price three months after the end of the chart and to report their *forecast confidence*, the probability that the realized price falls within five points of that forecast. We report the complement of forecast confidence as *forecast uncertainty*, so that higher values indicate greater perceived risk. Finally, participants make an *investment* decision by allocating \$3 between the stock and a risk-free account. The fraction invested in the stock is the *investment share*, which allows us to assess how the other measures relate to investment choice. Participants report all four measures on a slider.

Incentives. Every question except the risk score is incentivized. Each participant’s bonus is determined by their response to one randomly selected question about a real chart.⁵ If the investment question is selected, the bonus equals the value of the \$3 endowment at the three-month horizon. If the forecast question is selected, the bonus is \$3 when the forecast falls within five points of the realized price and \$0 otherwise. If the confidence question is selected, we use a Becker–DeGroot–Marschak (BDM) mechanism. The computer draws a number uniformly between 1% and 100%. If the reported confidence is below that number, the participant receives the \$3 bonus with probability equal to the number; otherwise, the participant receives \$3 if the forecast falls within five points of the realized price. We tell participants that truthful and accurate reporting is in their best interest and make the details of the mechanism available without requiring participants to study them.⁶

⁵Selecting one question at random encourages participants to treat each question in isolation and removes the interdependence between answers that income effects and hedging motives would otherwise create (Azrieli et al., 2018). Paying on a real chart has two further benefits. First, it frees participants from second-guessing how we would determine future prices for the synthetic charts. Second, it encourages them to treat every chart as real, which makes our findings more likely to be externally valid.

⁶Danz et al. (2022) show that participants report beliefs more truthfully when they are told that truthful reporting is in their best interest than when they are shown the quantitative details of the scoring rule.

Logistics. We recruit 502 participants on Prolific, all U.S. residents with at least 20 prior studies, a 99% approval rate, and self-reported stock-market experience. The median completion time is 22 minutes. Participants who complete the experiment receive a \$5 participation fee, and the mean total payment, including the bonus, is \$6.34. We pre-register the experiment on AsPredicted at https://aspredicted.org/Z7L_RRY.

II.B Choosing the outcome for feature selection

We use the data from the main experiment to build a model of risk perception from chart features. We fit the model to the risk score—the unincentivized but comprehensive measure—for two reasons. First, conceptually, the risk score is broader than forecast uncertainty: it allows participants to incorporate multiple dimensions of perceived risk rather than restricting them to a specific belief about the future return distribution. Selecting features on the risk score therefore makes us less likely to exclude features that matter for perceived risk but not for forecast uncertainty.

Second, empirically, the risk score is preferable for feature selection on two dimensions: it better explains the investment decision and contains more chart-level signal. Table 1 examines the first dimension by regressing the investment share on the belief measures and the risk score, with participant fixed effects and standard errors two-way clustered by participant and chart. The within-participant R^2 is 0.248 for the forecast and forecast uncertainty together, compared with 0.317 for the risk score alone and 0.403 when all three measures are included. In the joint regression, the risk score also has the largest standardized coefficient in absolute value ($-0.381, t = 23.0$).

The second dimension is chart-level signal: the extent to which responses vary systematically across charts. This matters for feature selection because the candidate predictors are chart features; an outcome with little systematic variation across charts provides little basis for distinguishing among competing feature models. Table 2 reports two measures of chart-level signal, together with the share of variation attributable to persistent differences across participants. Both measures of chart-level signal strongly favor the risk score. Chart fixed effects explain 24.8% of the within-participant variation in the risk score, compared with 4.5% for forecast uncertainty. The inter-rater

Table 1
Risk measures and investment allocation

This table reports regressions of the investment share on the risk score, the price forecast, and forecast uncertainty in the main experiment. The sample contains 10,040 responses from 502 participants, with each participant viewing 20 charts drawn from a pool of 200. The investment share is the fraction of a \$3 endowment allocated to the stock, and the risk score ranges from 0 to 10. Participants forecast the price three months after the chart ends. Forecast uncertainty equals 100 minus their reported probability, on a 0–100 scale, that the realized price will be within five points of the forecast. Column 1 includes the forecast and forecast uncertainty, column 2 includes the risk score, and column 3 includes all three measures. All variables are standardized, and all regressions include participant fixed effects. Standard errors are reported in parentheses and are two-way clustered by participant and chart. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Dependent variable: investment share	(1)	(2)	(3)
Risk score		−0.497*** (0.017)	−0.381*** (0.017)
Forecast	0.403*** (0.031)		0.266*** (0.025)
Forecast uncertainty	−0.256*** (0.022)		−0.144*** (0.018)
Participant fixed effects	Yes	Yes	Yes
Within-participant R^2	0.248	0.317	0.403
Observations	10,040	10,040	10,040

correlation is 0.176 for the risk score but only 0.008 for forecast uncertainty. Thus, two randomly drawn participants who view the same chart give risk scores with a correlation of about 0.18, whereas their forecast uncertainty reports are essentially uncorrelated. In addition, participant fixed effects absorb 67% of the raw variation in forecast uncertainty, indicating large and persistent differences in reported uncertainty across participants. We therefore use the risk score for feature selection and the incentivized belief measures to assess cross-measure performance and study the channels.

II.C Selecting a parsimonious feature model

The literature has identified many potential features of a price path that could affect risk perception, creating a “feature zoo.” Many of these features are empirically correlated or conceptually overlapping, which complicates interpretation and mechanism identification, and the large number of candidate features also raises concerns about overfitting.

Table 2
Variation in risk measures across charts and participants

This table compares variation in risk scores and forecast uncertainty across charts and participants in the main experiment. The sample contains 10,040 responses from 502 participants, with each participant viewing 20 charts drawn from a pool of 200. Risk scores range from 0 to 10. Forecast uncertainty equals 100 minus the reported probability that the realized price three months after the chart ends will fall within five points of the forecast. Column 1 reports the fraction of within-participant variation explained by chart fixed effects. Column 2 reports the inter-rater intraclass correlation (ICC), the correlation between the responses of two randomly drawn participants who view the same chart. Column 3 reports the fraction of variation in raw responses explained by participant fixed effects.

Measure (experiment)	Chart-FE R^2 (within-participant)	Inter-rater ICC	Participant-FE variance share
Risk score (main)	0.248	0.176	0.224
Forecast uncertainty (main)	0.045	0.008	0.673

We use the participant-reported risk score to select a parsimonious model based on three criteria: performance on held-out charts, performance across measures of perceived risk, and empirical and conceptual independence among the retained features. We start by assembling the candidate feature set from the prior literature.

Step 1: Feature collection from the literature. Borsboom and Zeisberger (2020) and Bose et al. (2022) each consider a large number of price-path features. We take the features from both papers that vary across single price paths, adapt their definitions to our setting, and include overlapping features only once. This yields a pool of 23 features—eleven from Tables 1 and 3 of Borsboom and Zeisberger (2020) and twelve from the feature table of the Internet Appendix of Bose et al. (2022). Table 3 lists these features and their definitions.

Step 2: Clustering. Since many of the 23 features are highly correlated, we group them using agglomerative hierarchical clustering with average linkage and $1 - |\rho|$ as the distance between two features, where ρ is their chart-level correlation. Two clusters merge when the average absolute correlation across all pairs of their members exceeds 0.5, yielding eleven clusters.⁷ Internet Appendix Table IA.1 reports the full merge sequence. For each cluster, we choose the medoid as its

⁷The clustering is robust to the threshold: the five features selected below remain in separate clusters down to a threshold of $|\rho| > 0.3$.

Table 3
Candidate price-path features and clusters

This table lists 23 candidate price-path features and groups them into eleven clusters. For each pair of features, we compute their Pearson correlation ρ across the 200 charts of the main experiment. Two clusters are merged when the average absolute correlation across all feature pairs from the two clusters is at least 0.5. Horizontal lines separate the resulting clusters. Within each cluster, the selected representative feature is shown in bold. The final column reports each feature’s average absolute correlation with the other members of its cluster. Superscript *a* marks features from Borsboom and Zeisberger (2020), and superscript *b* marks features from Bose et al. (2022); overlapping features are listed only once.

Feature	Definition	Mean $ \rho $
Volatility^a	Standard deviation of the return sequence, annualized	—
Volatility recency^a	Standard deviation of the last six monthly returns minus that of the first six	—
Price range^b	Highest minus lowest price on the price path	0.64
Mean distance from the start ^b	Mean absolute deviation of the price from its starting value	0.64
Total return^a	Final price divided by starting price, minus one	0.70
Average price ^b	Mean of the thirteen prices on the path	0.77
Maximum price ^a	Highest price on the path	0.73
Maximum crash ^a	Largest one-month fall in the price, in price points	0.69
Minimum price ^a	Lowest price on the path	0.69
Three-month trend ^a	Price in the final month minus the price three months earlier	0.57
Minimum return ^b	Most negative monthly return	0.54
Return autocorrelation^b	First-order autocorrelation of the return sequence	—
Gain-to-loss ratio^b	Months in which the price rose, divided by months in which it fell	0.76
Number of gains ^b	Number of months in which the price rose	0.73
Run length ^b	Average number of immediately preceding months whose return had the same sign	0.52
Maximum recovery^a	Largest one-month rise in the price, in price points	0.79
Maximum return ^b	Largest monthly return	0.78
Return skewness ^b	Skewness of the return sequence	0.68
Return kurtosis^a	Kurtosis of the return sequence	0.55
Jump ^b	Indicator for containing a monthly return in the top 5% of absolute returns across all charts	0.55
Months below start^a	Number of months whose price is below the starting price	—
Ever underwater^b	Indicator for the price ever falling below its starting value	—
Return difference between halves^a	Return over the last six months minus the return over the first six	—

representative, defined as the feature with the highest average absolute correlation with the other features in the cluster.⁸

We make one exception to the medoid rule. In the cluster containing total return, average price has the highest average absolute correlation with the other cluster members (0.77), followed by maximum price (0.73) and total return (0.70). We do not select average price or maximum price because both depend on which point of the price path is normalized to 100, whereas total return does not. We therefore select total return as the representative of this cluster.

Table 3 summarizes the clusters and their representatives and reports each feature’s average absolute correlation with the other features in its cluster.

Step 3: Selection by cross-validation. We then compare all 1,024 models that can be formed from the eleven cluster representatives while requiring volatility to be included. We rank these models by five-fold cross-validated R^2 , with folds formed over charts so that performance is evaluated on held-out charts. This procedure guards against selecting models that fit the estimation sample well but generalize poorly.⁹ Table 4 reports the cross-validated performance of the main candidate models. The highest cross-validated R^2 , 0.191, is achieved by a six-feature model including volatility, total return, volatility recency, return autocorrelation, price range, and the number of months the price spends below its starting price, which we call *months below start*.

The six-feature model performs best in cross-validation, but it does not fully satisfy our criterion of empirical independence. Table 5 shows that months below start is correlated at -0.80 with total return, whereas all other pairwise correlations among the six features are at most 0.46 in absolute value. This high correlation also makes the coefficient on months below start unstable. Its univariate coefficient is $+0.249$ (Table 6, column 1), but it reverses sign to -0.106 when all six features are included (column 3), while the other five features retain the same directional effects. Removing total return from the six-feature specification changes the coefficient on months below start back to

⁸Two clusters contain only two features, so the two members tie by construction; in these cases, we retain price range and return kurtosis, the more standard feature in each pair.

⁹Each candidate model is estimated by elastic net on the training folds, with the penalty and mixing parameter chosen by three-fold cross-validation within those folds.

Table 4
Cross-validated R^2 across price-path feature models

This table reports five-fold cross-validated R^2 for models of the risk score and forecast uncertainty in the main experiment. The sample contains 10,040 responses from 502 participants, with each participant viewing 20 charts drawn from a pool of 200. Each row corresponds to a feature set. The five-feature model includes volatility, volatility recency, price range, total return, and return autocorrelation; the six-feature model adds months below start. Table 3 lists the eleven cluster representatives and all 23 candidate features. The 200 charts are divided into five folds. In each outer fold, the model is estimated on four folds and evaluated on the remaining fold. Within the four training folds, three-fold cross-validation selects the elastic-net penalty and mixing parameter (0.5 or 0.9). The reported R^2 is calculated from the pooled outer-fold predictions. Entries are rounded to three decimals; in column 2, the five- and six-feature values are 0.0132 and 0.0128, respectively, to four decimal places.

Feature set	Risk score (main)	Forecast uncertainty (main)
	(1)	(2)
Volatility only	0.019	0.001
Five-feature (main)	0.187	0.013
Six-feature	0.191	0.013
Eleven cluster representatives	0.188	0.012
All 23 features	0.198	0.011

Table 5
Correlations among the six selected price-path features

This table reports Pearson correlations among the six price-path features across the 200 charts of the main experiment. Each feature is computed from a twelve-month price path whose final price is normalized to 100. Months below start is the number of month-end prices below the initial price; the other five features are defined in Table 3.

	Vol.	Recency	Range	Total ret.	Autocorr.	Months below
Volatility	1.00					
Recency	+0.04	1.00				
Range	+0.32	+0.00	1.00			
Total return	−0.15	−0.18	−0.46	1.00		
Autocorrelation	+0.10	+0.13	+0.36	−0.13	1.00	
Months below start	+0.13	+0.13	+0.39	− 0.80	+0.26	1.00

+0.128 (column 2).

These diagnostics lead us to drop months below start and retain the remaining five features as our preferred model. The loss in predictive performance is small: the cross-validated R^2 for the risk score declines from 0.191 to 0.187, and the five-feature model ranks second among the 210 volatility-including five-feature models that can be formed from the eleven representatives. Only the model that replaces volatility recency with months below start performs slightly better, with a

Table 6
Risk-score regressions with alternative feature sets

This table reports regressions of the risk score on alternative feature sets in the main experiment. The sample contains 10,040 responses from 502 participants, with each participant viewing 20 charts drawn from a pool of 200. Column 1 reports a separate regression for each feature, column 2 jointly includes the five features other than total return, and column 3 jointly includes all six features. The features are defined in Table 3. All variables are standardized, and all regressions include participant fixed effects. Standard errors are reported in parentheses and are two-way clustered by participant and chart. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Dependent variable: risk score	Each feature alone (1)	Five features, excluding total return (2)	All six features (3)
Volatility	0.139*** (0.025)	0.041** (0.016)	0.042*** (0.013)
Recency	0.110*** (0.032)	0.079*** (0.021)	0.043*** (0.017)
Range	0.315*** (0.025)	0.233*** (0.027)	0.162*** (0.021)
Total return	−0.333*** (0.027)		−0.316*** (0.035)
Autocorrelation	0.185*** (0.028)	0.048** (0.021)	0.099*** (0.019)
Months below start	0.249*** (0.027)	0.128*** (0.023)	−0.106*** (0.029)
Participant fixed effects	Yes	Yes	Yes
Within R^2		0.158	0.195
Observations	10,040	10,040	10,040

cross-validated R^2 of 0.190. Internet Appendix C shows that months below start largely reproduces the role of total return rather than providing an independent source of explanatory power.

Step 4: Cross-measure performance. We next assess whether the selected five features also explain forecast uncertainty, a narrower but incentivized measure of perceived risk elicited in the main experiment. Among all 1,024 volatility-including models built from the eleven cluster representatives, the selected five-feature model achieves the highest cross-validated R^2 for forecast uncertainty (0.013; Table 4) and also outperforms the full 23-feature model (0.011).

Taken together, the evidence on held-out performance, cross-measure performance, and empirical and conceptual independence leads us to adopt volatility, volatility recency, price range, total

return, and return autocorrelation as our main five-feature model.

III Channels and Mechanisms

Having selected the five features, we next ask why each affects perceived risk. We first examine whether each feature operates through forecast uncertainty or the perceived probability of a large downside forecast error. We then use targeted experiments to distinguish competing explanations for the total-return and volatility-recency effects and to separate the mechanisms behind the price-range and autocorrelation effects.

We conduct six additional experiments on independent samples of Prolific participants recruited under the same criteria as in the main experiment. Each retains the twenty-round format and three-month forecast horizon. Table 7 summarizes the participants, charts, elicited measures, and key design features of the main experiment and the six additional experiments. We describe each experiment in detail below.

III.A Does each feature operate through the perceived return distribution?

For each risk measure, Table 8 reports ordinary least squares (OLS) estimates of

$$Y_{ij} = \alpha_i + \sum_{k=1}^5 \beta_k X_{jk} + \varepsilon_{ij}, \quad (1)$$

where Y_{ij} is the risk measure reported by participant i for chart j , X_{jk} is feature k of chart j , and α_i is a participant fixed effect. The equation is estimated separately for each column of the table. The outcome and features are standardized within that column's estimation sample, and standard errors are two-way clustered by participant and chart.

Within the main experiment. We first examine forecast uncertainty, elicited alongside the risk score in the main experiment. Regressing forecast uncertainty on the five features shows that all five

Table 7
Experimental designs and samples

This table summarizes the main experiment and six additional experiments. Each participant completes 20 rounds, viewing one chart per round. N denotes the number of participants, with participant-round observations shown in parentheses. Measures are listed in the order in which they are elicited. The risk score is not incentivized; all other measures are incentivized. The main experiment uses 10 real and 10 reshuffled charts, while each additional experiment uses 12 real and 8 designed charts. The Position experiment includes 105 long investors and 93 short sellers.

Experiment	Elicited measures, in order	N (obs.)	Charts	Key design feature
<i>Main</i>	Risk score; forecast; forecast uncertainty; investment share	502 (10,040)	10 real, 10 reshuffled	The selection sample; the synthetic charts are random reshuffles of three fixed base sequences
<i>Uncertainty</i>	Forecast; forecast uncertainty	148 (2,960)	12 real, 8 designed	Forecast uncertainty is the only measure of risk perception elicited
<i>Downside</i>	Forecast; downside probability	150 (3,000)	12 real, 8 designed	Downside probability—the probability that the realized price falls more than 20 points below the participant's forecast—is the only risk perception measure elicited
<i>Position</i>	Forecast; risk score	198 (3,960)	12 real, 8 designed	A long or a short position, assigned at random at the start of the experiment
<i>Chosen-order</i>	Forecast; forecast uncertainty	150 (3,000)	12 real, 8 designed	The chart is concealed, and participants choose the order in which to reveal the twelve monthly returns, subject to a randomly assigned limit on the number of reveals
<i>Forced-order</i>	Forecast; forecast uncertainty	150 (3,000)	12 real, 8 designed	The chart is concealed and the computer reveals the twelve monthly returns one at a time in a random order
<i>Return-arrow</i>	Forecast; forecast uncertainty	150 (3,000)	12 real, 8 designed	The price line is replaced by the twelve monthly returns drawn as labeled arrows from a common zero line, with no price level shown

Table 8
Price-path features and measures of perceived risk

This table reports regressions of three risk-perception measures on the five selected price-path features. All regressions follow Equation (1). Columns 1 and 2 use the main experiment (502 participants, 10,040 observations), column 3 uses the Uncertainty experiment (148 participants, 2,960 observations), and column 4 uses the Downside experiment (150 participants, 3,000 observations). The five features are defined in Table 3. In the main experiment, forecast uncertainty is elicited after the risk score; in the Uncertainty and Downside experiments, the corresponding belief measure is the only risk-perception measure elicited. All variables are standardized, and all regressions include participant fixed effects. Standard errors are reported in parentheses and are two-way clustered by participant and chart. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

	Risk score	Forecast uncertainty		Downside
	(main)	(main)	(Uncertainty)	probability
	(1)	(2)	(3)	(4)
Volatility	0.042*** (0.013)	0.019** (0.008)	0.037** (0.017)	0.014 (0.011)
Recency	0.048*** (0.017)	0.023*** (0.008)	0.039** (0.016)	0.043*** (0.010)
Range	0.167*** (0.022)	0.022** (0.010)	0.097*** (0.023)	0.067*** (0.015)
Total return	−0.230*** (0.024)	−0.031*** (0.009)	0.004 (0.015)	0.009 (0.010)
Autocorrelation	0.080*** (0.018)	0.031*** (0.008)	0.048* (0.024)	0.020* (0.011)
Participant fixed effects	Yes	Yes	Yes	Yes
Within-participant R^2	0.191	0.017	0.061	0.038
Observations	10,040	10,040	2,960	3,000
Participants	502	502	148	150

effects are statistically significant and have the same signs as their counterparts for the risk score, but are uniformly smaller in magnitude (Table 8, column 2).

The Uncertainty experiment. Interpreting column 2 requires caution because forecast uncertainty is elicited after the risk score in every round of the main experiment. Answering a second risk-related question about the same chart may add noise, while participants may also anchor their uncertainty reports on the risk scores they have just given. We therefore run an additional experiment in which forecast uncertainty is the only measure of perceived risk elicited.¹⁰ This *Uncertainty experiment*

¹⁰Pre-registered on AsPredicted (#278198): <https://aspredicted.org/59hn5f.pdf>. The price path display is the same as in Figure 1, except that the price axis ranges from 50 to 150, all month-end prices are shown as integers and labeled, and participants enter their Month 3 forecast by clicking on the dashed vertical line.

($N = 148$) retains the twenty-round format, three-month forecast horizon, and BDM incentivization of the main experiment, with two main differences:

1. Participants report only a forecast and forecast confidence, with no risk score or investment decision. They enter the forecast by clicking on the dashed line at Month 3 and report forecast confidence—the probability that the realized price falls within five points of the forecast—on a slider.
2. Each participant sees twelve real charts from the same pool as in the main experiment and eight designed synthetic charts. The eight conditions are built from one of two base sequences and chosen so that pairs of charts differ mainly in one of the five features while holding the others as nearly fixed as possible (Internet Appendix Table IA.2). For example, the *Past* and *Recent* conditions reorder the same twelve monthly returns so that the largest moves occur early in one and late in the other, while leaving volatility, total return, return autocorrelation, and price range essentially unchanged. Since every participant sees both conditions, their difference provides a within-participant measure of the volatility-recency effect without relying on a fitted model. Internet Appendix B describes the full design and reports the corresponding comparisons.

The Downside experiment. A second risk-related aspect of the perceived return distribution is the probability that the realized price falls more than 20 points below the participant’s forecast, which we call the *downside probability*. We measure it in the *Downside experiment* ($N = 150$),¹¹ which is identical to the Uncertainty experiment except that the downside probability replaces forecast confidence.

Results in the two belief experiments. When elicited on their own, forecast uncertainty and the downside probability respond similarly to several of the five features (Table 8, columns 3 and 4). Volatility recency, price range, and autocorrelation affect both measures, while volatility affects

¹¹Pre-registered on AsPredicted (#279116): <https://aspredicted.org/n5pf5q.pdf>.

forecast uncertainty but not the downside probability.¹² Total return is the clear exception. Its estimates are close to zero and statistically insignificant in both experiments, in sharp contrast to its strong negative effect on the risk score.¹³ The model-free comparisons of the designed charts show a similar pattern (Internet Appendix Table IA.3).

III.B Total return: is it about personal loss?

The personal-loss hypothesis. The main experiment and the two belief experiments differ in participants' exposure to investment outcomes. In the main experiment, participants may invest up to \$3 in the stock and therefore face the prospect of gains or losses on a long position; in the Uncertainty and Downside experiments, they hold no position and are rewarded only for accurate beliefs. This difference may help explain the total-return effect. If the risk score partly reflects concern about personal investment losses, a history of falling prices may raise the risk score while having little effect on the two belief measures.

The Position experiment. To investigate this hypothesis, we run the *Position experiment*,¹⁴ in which we randomly assign each participant a long or a short position in the stock (198 participants; 105 long and 93 short). A future price decline therefore represents a loss for participants assigned to the long position but a gain for those assigned to the short position. The bonus is based on the value of the assigned position at the three-month horizon for a randomly selected real stock: it is proportional to the Month-3 price for the long position and to 200 minus that price for the short position. Participants report only a price forecast and the risk score, avoiding the additional complications of a confidence report and an investment decision. Each participant rates twenty charts: twelve real charts from the same pool of 100 as in the main experiment and eight designed charts as in the Uncertainty experiment (Internet Appendix B).

¹²The volatility coefficient in the Downside experiment is attenuated and statistically insignificant ($+0.014$, $p = 0.20$), consistent with the downside probability capturing asymmetry rather than dispersion.

¹³Months below start, which is correlated with total return at -0.80 and was dropped in Step 3 of Section II.C, shows the same pattern when substituted for total return: it affects the risk score but not forecast uncertainty or the downside probability. Internet Appendix Table IA.5 reports these estimates and additional evidence separating the two features.

¹⁴Pre-registered on AsPredicted (#282827): <https://aspredicted.org/ek2ky3.pdf>.

Risk scores by position. Panel A of Table 9 shows no statistically significant difference in average risk scores between the long and short groups. To compare feature effects across positions, Panel B estimates a pooled five-feature regression with interactions between each feature and a short-position indicator. The long coefficients are the main effects, the short coefficients are the main effects plus interactions, and the differences are the interaction coefficients. The total-return coefficient is negative in both groups, at -0.131 for the long group and -0.074 for the short group. The short-minus-long difference in the total-return coefficient is not statistically significant ($p = 0.13$), and a joint test also fails to reject equality of all five feature coefficients across positions ($p = 0.35$). Pairwise comparisons of the designed charts show the same pattern (Internet Appendix Table IA.4).

Interpreting the results. The short group responds less strongly to the chart features overall: the five-feature model explains 15.2% of within-participant variation in risk scores in the short group, compared with 21.2% in the long group. The short group also shows smaller responses in each of the four designed-chart comparisons in Internet Appendix Table IA.4, including comparisons unrelated to gains or losses. The smaller magnitude of the total-return coefficient in the short group may therefore reflect this broader attenuation rather than a response specific to personal gains and losses.¹⁵

Taken together, total return strongly affects risk scores, has little effect on the two separately elicited belief measures, and continues to affect risk scores among participants with a short position. These findings are difficult to reconcile with an explanation based solely on personal investment losses. One possibility is that participants draw inferences about the stock's fundamentals from its past return.

¹⁵The same pattern holds when the direction of the price path is measured by months below start. Replacing total return with months below start yields positive coefficients in both the long and short groups, with no statistically significant difference between the two coefficients (Internet Appendix Table IA.6).

Table 9
Risk scores under randomly assigned long and short positions

This table reports risk-score regressions in the Position experiment, with 3,960 responses from 105 participants in the long-position group and 93 in the short-position group. Participants are randomly assigned a long or short position before reporting risk scores. Panel A regresses the risk score on a short-position indicator with chart fixed effects; the coefficient measures the average difference between the short and long groups. Panel B pools both groups and estimates the following extension of Equation (1):

$$Y_{ij} = \alpha_i + \sum_{k=1}^5 \beta_k X_{jk} + \sum_{k=1}^5 \delta_k D_i X_{jk} + \varepsilon_{ij}.$$

where D_i equals one for a short position and zero for a long position. The main effect of D_i is absorbed by participant fixed effects. Column 1 reports β_k , the feature coefficients for the long group; column 2 reports $\beta_k + \delta_k$, the corresponding coefficients for the short group; and column 3 reports δ_k , the short-minus-long differences in the feature coefficients. The within-participant R^2 values are calculated from separate five-feature regressions within the long and short groups. The five features are defined in Table 3. All nonindicator variables are standardized. Standard errors are reported in parentheses and are two-way clustered by participant and chart. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Dependent variable: risk score	Long position (1)	Short position (2)	Difference (3)
<i>Panel A. Average level of perceived risk</i>			
Average risk score			−0.007 (0.083)
<i>Panel B. The five-feature model</i>			
Volatility	0.071** (0.030)	0.039** (0.020)	−0.032 (0.024)
Recency	0.092*** (0.030)	0.070*** (0.026)	−0.022 (0.020)
Range	0.227*** (0.034)	0.232*** (0.034)	0.005 (0.039)
Total return	−0.131*** (0.029)	−0.074** (0.031)	0.056 (0.037)
Autocorrelation	0.068** (0.029)	0.053** (0.027)	−0.014 (0.030)
Participant fixed effects (Panel B)	Yes	Yes	Yes
Within-participant R^2 (Panel B)	0.212	0.152	
Observations	2,100	1,860	3,960
Participants	105	93	198

III.C Volatility recency: perceived relevance or viewing order?

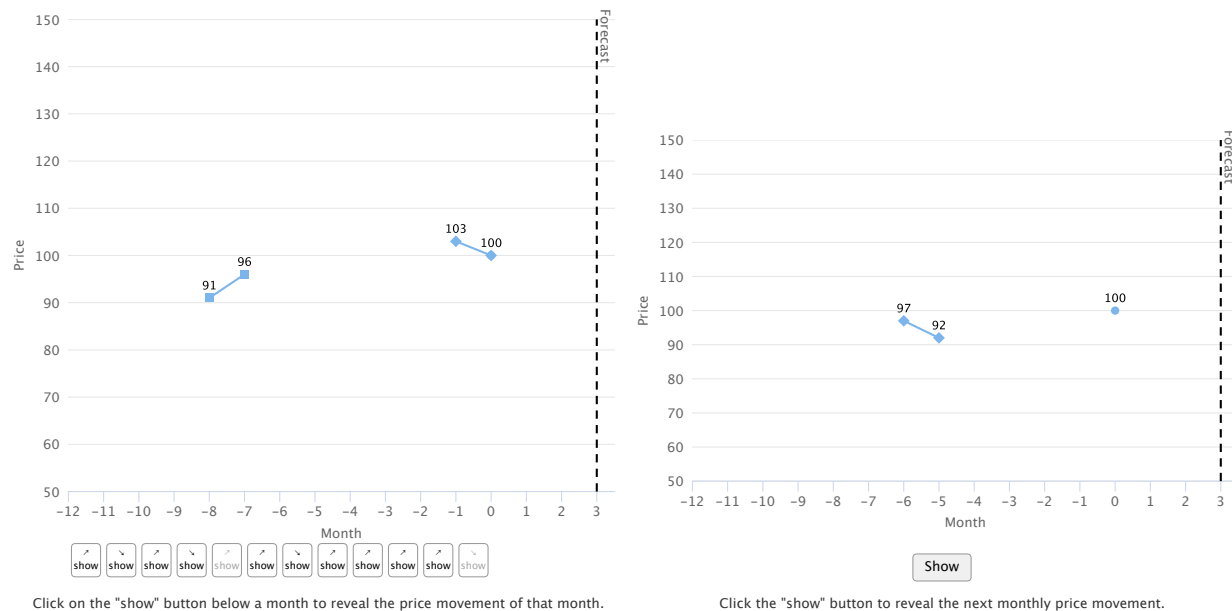
The three order-related features, volatility recency, price range, and autocorrelation, affect forecast uncertainty when it is elicited on its own. We therefore continue to use forecast uncertainty as the outcome in the diagnostic experiments. The Uncertainty experiment serves as the benchmark because it uses the same designed charts and presents the full price history at once, without manipulating how participants view it.

Two competing explanations. We first focus on volatility recency. Participants may give greater weight to recent volatility because they consider it more informative about future risk. Alternatively, recent returns may receive greater weight simply because they are viewed first. We distinguish these explanations with two experiments. The Chosen-order experiment examines which returns participants choose to inspect first, while the Forced-order experiment randomizes reveal order to test whether seeing a return earlier increases the weight it receives.

The Chosen-order experiment. We run the *Chosen-order experiment*, in which the chart is initially concealed and participants reveal the twelve monthly returns one by one in an order of their choice (Figure 2a).¹⁶ Before each reveal, participants see the month to which the return belongs and whether the price rose or fell in that month. They can therefore choose which returns to inspect based on calendar position, return sign, streaks of same-signed returns, and turning points in the price path.

To make reveal order incentive-relevant, participants face a randomly assigned cap on the number of returns they may reveal before reporting their forecast and forecast confidence. This gives them an incentive to reveal the returns they consider most informative first. Each participant faces one round with each cap from one through five reveals, five rounds capped at six reveals, and ten uncapped rounds. Participants cannot report their forecast before reaching the cap or, in an uncapped round, before revealing all twelve returns.

¹⁶Pre-registered on AsPredicted (#279392): <https://aspredicted.org/js9p4z.pdf>.



(a) Chosen-order.

(b) Forced-order.

Figure 2. Price-chart displays in the Chosen-order and Forced-order experiments.

This figure shows the displays used in the Chosen-order and Forced-order experiments. Both initially show only the final price, normalized to 100 at Month 0, on a price axis from 50 to 150. Participants enter their Month-3 forecast on the dashed vertical line. In panel (a), twelve buttons allow participants to choose which monthly price movement to reveal next. Each button shows the movement's direction before reveal, while its size remains hidden. Used buttons turn gray; two movements have been revealed in the example. Participants respond after reaching the randomly assigned reveal limit or, in uncapped rounds, after revealing all twelve movements. In panel (b), a single button reveals movements in a computer-chosen random order. Each new movement replaces the previous one, so only one is visible at a time; there are no direction arrows or separate month buttons. Participants respond after all twelve movements have been shown.

Alt text: Two experimental price-chart interfaces. Panel (a) shows twelve month-specific reveal buttons marked with upward or downward arrows; two price movements are visible and the corresponding buttons are gray. Panel (b) shows one visible price movement and a single Show button for computer-ordered reveals. Both panels end at a price of 100 at Month 0 and mark Month 3 with a dashed forecast line.

Table 10
Return features and reveal position in the Chosen-order experiment

This table relates the features of revealed returns to their reveal positions in the Chosen-order experiment (150 participants). Each column regresses a return feature on reveal position, with fixed effects for each participant-chart round; an observation is one reveal event. Panel A uses rounds with at least two reveals, and Panel B uses uncapped rounds in which all twelve returns were revealed. Column 1 uses the calendar month of the revealed return as the dependent variable, numbered from 1 (earliest) to 12 (latest). Column 2 uses an indicator for a nonnegative return; rounding displayed prices to integers can produce zero returns. Columns 3 and 4 use indicators for whether either endpoint of the revealed return is a peak or trough, respectively. A price is a peak if it is at least as high as the preceding price and higher than the following price, and a trough if it is lower than the preceding price and no higher than the following price; the first and last prices are excluded. Columns 5 and 6 use the corresponding peak and trough indicators weighted by one plus the number of consecutive months without another turning point on either side. Each coefficient measures the change in the dependent variable associated with a one-position-later reveal; for the binary outcomes in columns 2–4, the coefficients are changes in probability. Standard errors are reported in parentheses and are two-way clustered by participant and chart. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

	Dependent variable: a feature of the return revealed					
			Return is at a turning point			
			Indicator		Weighted by isolation	
	Month	Positive	Peak	Trough	Peak	Trough
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. All reveal events</i>						
Reveal position	−0.240*** (0.047)	−0.004* (0.002)	0.001 (0.002)	0.008*** (0.002)	−0.007* (0.004)	0.019*** (0.005)
Reveal events	24,600 (2,850 rounds)					
<i>Panel B. Rounds in which all twelve returns were revealed</i>						
Reveal position	−0.232*** (0.047)	−0.003* (0.002)	0.003** (0.002)	0.004** (0.002)	−0.001 (0.003)	0.003 (0.002)
Reveal events	18,000 (1,500 rounds)					
Round fixed effects	Yes					
Participants	150					

What participants reveal first. Reveal order strongly tracks recency (Table 10). To characterize which returns participants choose to inspect earlier, we treat each reveal as an observation and regress a feature of the revealed return on its position in the reveal sequence, with round fixed effects. The coefficients therefore capture the within-round relationship between when a return is revealed and what kind of return it is.¹⁷

Later reveal positions correspond to returns further back in the window: each additional position is associated with a return 0.240 months more distant from the present, and the estimate is nearly identical in uncapped rounds (0.232). Individual reveal sequences show the same pattern more directly. In 34% of rounds with at least three reveals, participants reveal the three most recent returns first and in recency order; 51 of the 150 participants follow this recent-first pattern in a majority of their ten uncapped rounds. The reverse, distant-to-recent pattern is much less common (14% of rounds; 19 participants). Return sign and turning points are also related to reveal order, but less strongly: positive returns tend to be revealed earlier, while returns at troughs tend to be revealed later.¹⁸ Since the cap gives participants an incentive to reveal the returns they consider most informative first, the strong recent-first pattern suggests that they assign greater relevance to recent returns.

The Forced-order experiment. The Chosen-order experiment shows which returns participants choose to inspect first, but it leaves open whether viewing a return earlier causes it to receive greater weight. Bohren et al. (2025) show that limited attention and memory affect the weight placed on salient outcomes in risky choice, depending on whether outcomes are observed all at once or

¹⁷We place reveal position on the right-hand side because the randomly assigned cap censors reveal ranks. In the conventional specification, with reveal rank as the dependent variable, unrevealed returns have missing ranks precisely because their ranks exceed the cap. In our specification, censoring instead selects observations on the regressor. Since the cap is randomly assigned, this leaves the least-squares slope unbiased when the expected feature value is linear in reveal position; if it is not, the caps place greater weight on earlier positions. Panel B of Table 10 therefore also reports the uncapped rounds, where no censoring occurs. On this subsample, the two specifications use the same observations and give the same coefficient for the month feature.

¹⁸We measure turning points in two ways. The first is an indicator: a return spans two month-end prices and is classified as being at a peak if either price is higher than the prices immediately before and after it, and at a trough if either is lower. The second weights this indicator by one plus the number of consecutive months without another turning point on either side. The two measures give qualitatively similar results.

sequentially. To test whether viewing order matters here, we conduct the *Forced-order experiment*,¹⁹ in which participants reveal the twelve monthly returns one at a time in a computer-chosen random order (Figure 2b) and report their forecast and forecast confidence only after all twelve have been shown. When a new return appears, the previous one disappears, so only one return is visible at a time. If viewing order itself drives the recency effect, randomizing reveal order should weaken the effect of calendar recency and generate an effect of reveal order instead.

Results under random order. Randomizing reveal order leaves the relationship between forecast uncertainty and the five features essentially unchanged (Table 11, Panel A). None of the five coefficients differs significantly between the Forced-order and Uncertainty experiments, and a joint test does not reject equality of the five coefficients across the two experiments ($p = 0.24$). In particular, the volatility-recency coefficient remains positive and similar in magnitude (+0.031 in the Forced-order experiment versus +0.039 in the Uncertainty experiment). Thus, the greater weight on recent volatility does not depend on participants being free to view recent returns first.

Panel B tests more directly whether returns viewed earlier receive greater weight. We split the twelve returns into the six revealed first and the six revealed last and estimate the effect of the volatility of each group on forecast uncertainty. The two coefficients are similar: +0.035 for the first six and +0.027 for the last six, with a statistically insignificant difference ($p = 0.48$). Thus, reveal order itself has no detectable effect on forecast uncertainty.

III.D Price range and autocorrelation: one mechanism or two?

The Return-arrow experiment. We next turn to the other two order-related features, price range and return autocorrelation. The two are correlated: charts with many consecutive same-signed returns tend to have wider price ranges ($\rho = 0.36$ across the 200 charts of the main experiment). This raises the question of whether the two effects reflect the same mechanism. Conceptually, however, they differ: autocorrelation is a feature of the *return sequence*, whereas price range is

¹⁹Pre-registered on AsPredicted (#279389): <https://aspredicted.org/r4wb8d.pdf>.

Table 11**Forecast uncertainty under randomized reveal order in the Forced-order experiment**

This table compares forecast-uncertainty regressions in the Uncertainty experiment (148 participants, 2,960 observations) and the Forced-order experiment (150 participants, 3,000 observations). The Uncertainty experiment shows the full price chart at once, whereas the Forced-order experiment reveals all twelve monthly returns one at a time in a computer-chosen random order before participants report their forecasts. Panel A pools the two experiments and estimates the following extension of Equation (1):

$$Y_{ij} = \alpha_i + \sum_{k=1}^5 \beta_k X_{jk} + \sum_{k=1}^5 \delta_k F_i X_{jk} + \varepsilon_{ij}.$$

where F_i equals one for the Forced-order experiment and zero for the Uncertainty experiment. The main effect of F_i is absorbed by participant fixed effects. Column 1 reports β_k , the feature coefficients in the Uncertainty experiment; column 2 reports $\beta_k + \delta_k$, the corresponding coefficients in the Forced-order experiment; and column 3 reports δ_k , the Forced-order-minus-Uncertainty differences in the feature coefficients. Panel B uses only the Forced-order experiment. Within each round, the twelve returns are split into the six revealed first and the six revealed last. The standard deviation of each group enters the same regression, together with the other four features and participant fixed effects. The final row reports the coefficient on the volatility of the last six returns minus the coefficient on the volatility of the first six. The five features are defined in Table 3. Forecast uncertainty is defined as in Table 8. All nonindicator variables are standardized. Standard errors are reported in parentheses and are two-way clustered by participant and chart. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Dependent variable: forecast uncertainty	Uncertainty (1)	Forced order (2)	Difference (3)
<i>Panel A. The five calendar features</i>			
Volatility	0.037** (0.017)	0.050*** (0.009)	0.013 (0.020)
Recency	0.039** (0.016)	0.031** (0.014)	−0.008 (0.012)
Range	0.098*** (0.024)	0.078*** (0.017)	−0.020 (0.027)
Total return	0.004 (0.015)	0.032** (0.014)	0.029 (0.019)
Autocorrelation	0.048* (0.024)	0.033** (0.013)	−0.015 (0.027)
<i>Panel B. Reveal order (Forced-order experiment)</i>			
Volatility of the six returns revealed first		0.035*** (0.008)	
Volatility of the six returns revealed last		0.027*** (0.009)	
Difference in coefficients (last − first)		−0.008 (0.011)	
Participant fixed effects	Yes	Yes	Yes
Observations	2,960	3,000	5,960
Participants	148	150	298

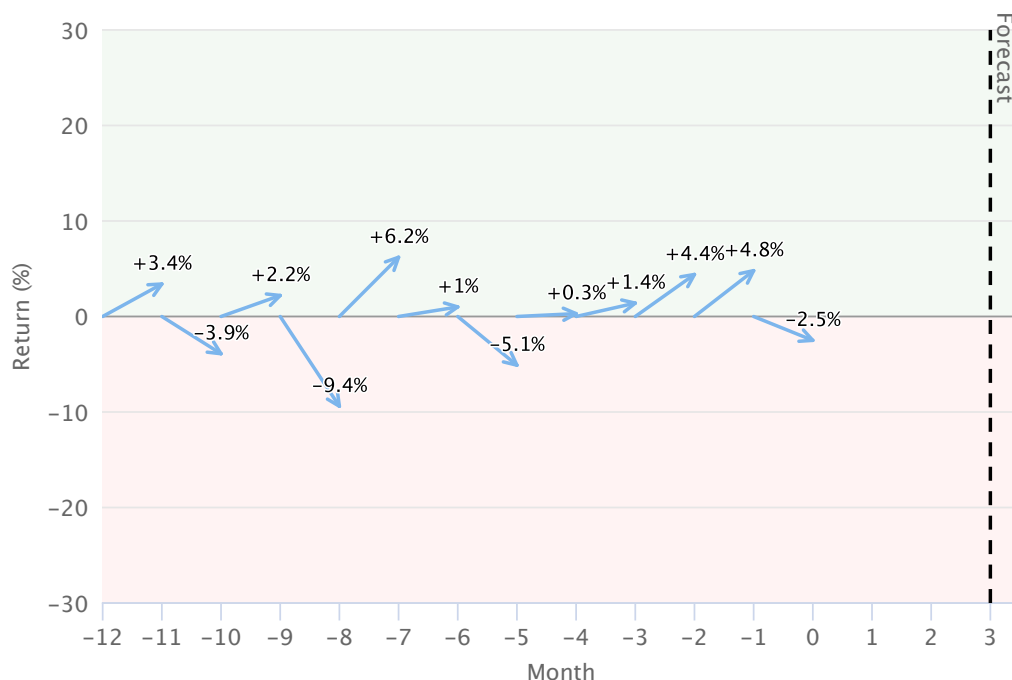


Figure 3. Monthly-return display in the Return-arrow experiment.

This figure shows the Return-arrow display for the price path in Figure 1. Each of the twelve monthly returns is shown as a labeled arrow from the zero line to its return value. The regions above and below zero are shaded green and red, respectively. The return axis ranges from -30 to 30 percent, and no price levels are shown. Participants enter their Month-3 forecast on the dashed vertical line as a percentage change from Month 0. Since the Month-0 price is normalized to 100, the forecast price equals 100 plus this percentage change.

Alt text: Display of twelve monthly returns as labeled arrows from a zero line, with positive returns extending upward and negative returns downward. The region above zero is shaded green and the region below zero is shaded red. A dashed line at Month 3 marks the forecast entry point, and no price path or price levels are shown.

a feature of the *price path*. Glaser et al. (2019) show that presenting information in prices rather than returns can affect expectation formation. To distinguish the two mechanisms, we run the *Return-arrow experiment*,²⁰ which replaces the price line with monthly returns drawn as arrows from a common zero line and labeled with their values (Figure 3). The return-arrow display removes the visual price path while preserving the sequence of returns. If the range effect depends on the price path but the autocorrelation effect depends on return ordering, the former should weaken while the latter should remain.

²⁰Pre-registered on AsPredicted (#283393): <https://aspredicted.org/3g2m8v.pdf>.

Table 12
Forecast uncertainty under price-line and return-arrow displays

This table compares forecast-uncertainty regressions under the price-line display in the Uncertainty experiment (148 participants, 2,960 observations) and the Return-arrow display (150 participants, 3,000 observations). The price-line display shows the full price path, whereas the Return-arrow display replaces the price line with twelve labeled monthly-return arrows drawn from a common zero line, with no price levels shown. The table pools the two experiments and estimates the following extension of Equation (1):

$$Y_{ij} = \alpha_i + \sum_{k=1}^5 \beta_k X_{jk} + \sum_{k=1}^5 \delta_k Z_i X_{jk} + \varepsilon_{ij}.$$

where Z_i equals one for the Return-arrow display and zero for the price-line display. The main effect of Z_i is absorbed by participant fixed effects. Column 1 reports β_k , the feature coefficients under the price-line display; column 2 reports $\beta_k + \delta_k$, the corresponding coefficients under the Return-arrow display; and column 3 reports δ_k , the Return-arrow-minus-price-line differences in the feature coefficients. Only the coefficients on price range and return autocorrelation are shown; volatility, volatility recency, and total return are included in the regression but omitted from the table. The five features are defined in Table 3. Forecast uncertainty is defined as in Table 8. All nonindicator variables are standardized. Standard errors are reported in parentheses and are two-way clustered by participant and chart. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Dependent variable: forecast uncertainty	Price line (1)	Return arrows (2)	Difference (2)–(1)
Range	0.097*** (0.023)	0.011 (0.018)	–0.086*** (0.025)
Autocorrelation	0.048* (0.024)	0.053*** (0.017)	0.005 (0.024)
Observations (pooled)	5,960		
Participants	298		

Results with return arrows. Changing the display separates the two effects (Table 12). The range coefficient falls from +0.097 with the price-line display to +0.011 with the return-arrow display, a significant decline ($p = 0.001$). By contrast, the autocorrelation coefficient is essentially unchanged (+0.048 versus +0.053; $p = 0.84$ for the difference). These results indicate that the two effects arise through different mechanisms: the range effect depends on seeing the price path, whereas the autocorrelation effect survives when price levels are removed.

IV Risk Perception and Market Outcomes

This section asks whether the five-feature model constructed using experimental data helps explain risk-related market outcomes. Holding the weights of the five-feature model at their experimental

values, we examine two questions. First, we assess whether an index built from the model behaves like a market measure of risk by comparing it with option-implied volatility and testing whether it predicts subsequent realized volatility in optionable U.S. stocks. Second, we examine whether market demand responds to the index in the same direction as participants’ investment choices in the experiment. In the main experiment, participants invested less in stocks whose charts they perceived as riskier (Table 1), so we test whether higher index values are associated with lower demand among retail investors and mutual-fund shareholders.²¹ We first construct the indices and describe the stock data, then report results setting by setting.

IV.A Two indices based on the five-feature model

Feature construction. To adapt the five features to market settings, we make minimal modifications to their definitions. All five features are constructed from an asset’s trailing monthly returns. Volatility, volatility recency, total return, and return autocorrelation follow the experimental formulas exactly. Price range requires one modification because its value depends on how the price chart is scaled. Since we do not observe the scaling investors face in market data, we normalize the cumulative gross-return path to one at the start of the window and measure range as its maximum minus its minimum, including the starting value. Before constructing the indices, we standardize each feature as a z -score.²² Let z_k denote the standardized value of feature k for a given price path, and let V , R , G , D , and A denote volatility, volatility recency, price range, total return, and return autocorrelation, respectively.

The chart-risk index. Let $\hat{\beta}_k$ denote the coefficient on feature k from the main experiment’s risk-score regression estimated using the market feature definitions (Internet Appendix Table IA.10).

²¹ Market return paths are not randomized, so we do not infer causality from these results.

²² For stocks, observations in year y are standardized using stock observations from 2001 through $y - 1$; for mutual funds, we use all share-class months from 2000 onward for which all five features are observed. The exact feature definitions and standardization procedures are described in Internet Appendix D.

We define the *chart-risk index* as

$$S^E = \frac{\sum_{k \in \{V, R, G, D, A\}} \hat{\beta}_k z_k}{\sigma_E}, \quad (2)$$

where σ_E is the standard deviation of $\sum_{k \in \{V, R, G, D, A\}} \hat{\beta}_k z_k$ across the stock or fund observations used to standardize the features. Equation (2) therefore combines all five features using their experimental risk-score weights and scales the resulting index to have unit standard deviation. The weights are estimated entirely from the experimental data; no market outcome enters their estimation. Total return and price range carry most of the experimental weight, while volatility carries relatively little.²³

The sequence index. To isolate the order-related content of the price path, we construct the *sequence index* using only volatility recency, price range, and return autocorrelation:

$$S^S = \frac{\sum_{k \in \{R, G, A\}} \hat{\beta}_k z_k}{\sigma_S}, \quad (3)$$

where σ_S is the standard deviation of $\sum_{k \in \{R, G, A\}} \hat{\beta}_k z_k$ across the stock or fund observations used to standardize the features. Equation (3) uses the same experimental weights as Equation (2) but retains only the three features that can change when the same returns are reordered. In regressions using S^S , we also control for volatility and total return. The coefficient on S^S therefore measures the association with the order-related content of the path, holding fixed the two features that are invariant to return order.

²³An index constructed using the weights from the incentivized forecast-uncertainty measure, S^C , correlates 0.89 with S^E . Replacing S^E with S^C in the regressions for option-implied volatility, future realized volatility with and without controlling for current implied volatility, and retail net selling in the high-share and full samples preserves both the sign and statistical significance of every coefficient (Internet Appendix Table IA.13).

IV.B Stock data and specification

For each stock-month, we construct the two indices from the stock’s trailing twelve-month price path and compare them with option-implied volatility, the market’s forward-looking measure of risk.

Data. The stock panel covers CRSP common stocks on the NYSE, Amex, and Nasdaq with listed single-name options, subject to standard price and size screens. The resulting sample contains 3,351 stocks from January 2001 through August 2025, the span of the option data. The optionability requirement accounts for most of the sample reduction, reducing the screened CRSP universe from roughly ten thousand stocks to about four thousand before imposing the twelve-return requirement (Internet Appendix D). The estimates therefore describe the optionable cross-section.

Windows and standardization. We construct the five features from twelve monthly returns in months $t - 11$ through t , matching the experimental horizon; 6- and 24-month windows serve as robustness checks. Since the features are measured in different units, we standardize them before applying the experimental weights. For each test year, the standardizing means and standard deviations are estimated using all stock-months from prior years. Thus, the 2006 indices use 2001–2005 data, the 2007 indices use 2001–2006 data, and so on. We scale each weighted index by its standard deviation in the same reference sample, so that one unit corresponds to one standard deviation based only on prior data. We reserve 2001–2005 as the initial reference period, so the indices begin in 2006 (Internet Appendix D).

Outcome variables and specification. The main outcome is log 30-day model-free implied volatility, computed from OptionMetrics option quotes (Internet Appendix D). We regress the outcome on either one of the two indices or the five features jointly, with stock and calendar-month fixed effects and previous-December log market capitalization as a control. Standard errors are two-

way clustered by stock and month.²⁴ Internet Appendix Table IA.8 gives the outcome definitions and timing conventions, and Table IA.9 reports summary statistics.

IV.C Implied and realized volatility

Time-series comovement. We begin by comparing the two indices with option-implied volatility, the market’s forward-looking measure of risk. Figure 4 plots the monthly cross-sectional averages of the two indices against average log implied volatility. The chart-risk index peaks in late 2008 and rises sharply in March 2020, with a correlation of 0.666 with mean log implied volatility (Panel (a)). The main divergence occurs in 2021, when strong post-COVID returns lower the index because total return enters with a large negative weight. The sequence index also tracks implied volatility, with a correlation of 0.533 (Panel (b)).

Contemporaneous implied volatility. Table 13 presents the regression evidence. A one-standard-deviation increase in the chart-risk index S^E is associated with 0.15 standard deviations higher log implied volatility, or about 7 percent (column 1). The sequence index is also positively associated with implied volatility after controlling for volatility and total return (column 2). Both results are robust to using 6- and 24-month price-path windows (Internet Appendix Table IA.12, Panel A).

When the five features enter jointly, each is significantly related to option-implied volatility with the sign predicted by the experiment (Table 13, column 3).²⁵ The two sets of coefficients agree in sign, with a cosine similarity of 0.905 (Internet Appendix E). The main difference is that implied volatility places much greater weight on volatility itself, consistent with its focus on expected variance rather than a broader judgment of risk. The three order-related features have positive and significant coefficients in both the experiment and the option market.²⁶

²⁴Index regressions begin in 2006 because the expanding standardization requires prior data. The five-feature regressions instead use the full 2001–2025 sample, with each feature standardized within the estimation sample.

²⁵The results are similar at the 6- and 24-month windows: all five coefficients have the experimental signs, and all but one are statistically significant (Internet Appendix Table IA.12, Panel B).

²⁶As a robustness check, we re-estimate the option-market coefficients separately for each test year, using only observations from preceding years. Volatility, recency, range, and total return retain their experimental signs in all 20 test years, while autocorrelation does so in 14 years (Internet Appendix E).

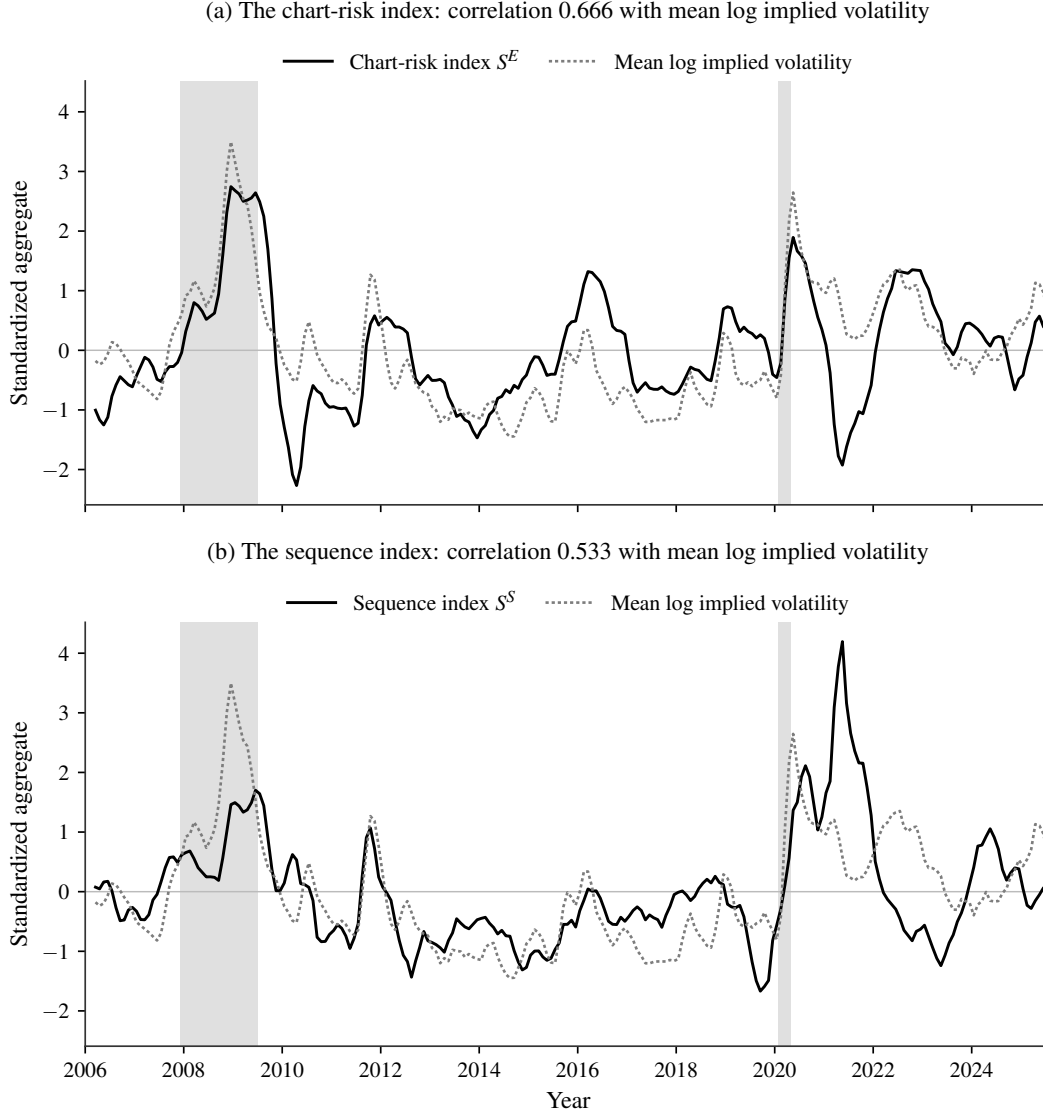


Figure 4. Chart-risk and sequence indices alongside implied volatility.

This figure plots monthly cross-sectional averages of the chart-risk index S^E and the sequence index S^S , defined in Equations (2) and (3), alongside mean log implied volatility for optionable CRSP stocks over 2006–2025. Panel (a) shows S^E , which uses all five features, and panel (b) shows S^S , which uses volatility recency, price range, and return autocorrelation. Features are computed from twelve monthly returns ending in the observation month. For each stock-month, daily 30-day model-free implied volatility is averaged within the month and then logged. For each month, the three series are equally weighted across stocks; the resulting monthly series are standardized and then smoothed using a trailing three-month moving average. The correlations in the panel headings are calculated from the unsmoothed series. Solid black lines show the indices, dotted gray lines show mean log implied volatility, and shaded bands mark NBER recessions.

Alt text: Two time-series panels compare the standardized indices with mean log implied volatility from 2006 to 2025. In panel (a), the chart-risk index and implied volatility generally move together, with pronounced peaks in 2008 and 2020, followed by a notable divergence in 2021; their unsmoothed correlation is 0.666. In panel (b), the sequence index also broadly co-moves with implied volatility, including sharp increases in 2008 and 2020, with an unsmoothed correlation of 0.533.

Table 13
Price paths and option-implied volatility

This table reports regressions of log implied volatility on price-path indices and individual price-path features for optionable CRSP common stocks. Log implied volatility is the log of the month- t average of daily 30-day model-free implied volatility from OptionMetrics and is standardized. The features are computed from twelve monthly returns ending in t . Column 1 regresses log implied volatility on the chart-risk index S^E . Column 2 regresses log implied volatility on the sequence index S^S together with volatility and total return. Column 3 regresses log implied volatility jointly on all five individual features. The indices are defined in Equations (2) and (3), and the market feature definitions are reported in Internet Appendix Table IA.7. Columns 1 and 2 cover 2006–2025, when the indices are available; column 3 covers 2001–2025. All regressions include stock and calendar-month fixed effects and control for log market capitalization at the end of the previous December. Standard errors are reported in parentheses and are two-way clustered by stock and month. For the coefficients on S^E and S^S , standard errors also incorporate uncertainty in the estimated experimental weights using the delta method. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

	Chart-risk index (1)	Sequence index with V and D (2)	Five features (3)
Volatility (V)		0.1728*** (0.0408)	0.1433*** (0.0170)
Recency (R)			0.0707*** (0.0154)
Range (G)			0.2813*** (0.0388)
Total return (D)		−0.1771*** (0.0101)	−0.2946*** (0.0250)
Autocorrelation (A)			0.0077*** (0.0028)
Chart-risk index S^E	0.1515*** (0.0208)		
Sequence index S^S		0.1389*** (0.0269)	
Observations	181,638	181,638	192,814

Subsequent implied and realized volatility. The contemporaneous regressions leave open whether the indices contain information about future risk. Table 14 therefore turns to outcomes measured after the feature window. Using a path that ends in month $t - 1$, both indices predict month- t implied volatility after controlling for month- $(t - 1)$ implied volatility (column 1); a one-standard-deviation increase in either index predicts about 0.04 standard deviations higher implied volatility. Both indices predict realized volatility over the following year after controlling for current implied volatility (Panel B).²⁷ When the five features enter jointly, volatility recency, price range, and total return retain the experimental sign and are significant across the forward outcomes; volatility predicts next-month implied volatility but not realized volatility, while autocorrelation predicts neither (Panel A).

Overall, the experimental indices carry over beyond the laboratory and contain information about future realized volatility beyond current implied volatility.

IV.D Retail selling

We next turn from market measures of risk to investor demand. In the main experiment, participants invest less in stocks they perceive as riskier. We examine the corresponding demand response in two market settings, beginning with retail investors and then turning to mutual funds in Section IV.E.

Data and specification. We use the twelve-month stock indices constructed in Section IV.B and measure retail order flow for optionable stocks from 2015 through 2023 using the subpenny classification of Boehmer et al. (2021). The outcome is retail net selling over months $t + 1$ through $t + 12$, defined as sell minus buy dollars divided by their sum. Regressions control for size, the month- t return, and trailing retail imbalance through month t . The latter two controls account for the contrarian tendency of retail trading documented by Kaniel et al. (2008) and for persistence in

²⁷To assess how much explanatory power is lost by imposing the experimental weighting, Internet Appendix Table IA.11 compares the restricted model, which combines recency, range, and autocorrelation into S^S using their experimentally estimated risk-score coefficients while estimating separate coefficients on volatility and total return, with the unrestricted model, which estimates separate coefficients on all five features. The restricted model's within- R^2 is 88.1% of the unrestricted model's within- R^2 for implied volatility and 69.4% for future realized volatility.

Table 14
Price paths and subsequent volatility

This table reports regressions of subsequent implied and realized volatility on price-path indices and individual price-path features for optionable CRSP stocks. Column 1 uses month- t log implied volatility as the outcome and constructs the price path from the twelve monthly returns ending in $t-1$; lagged log implied volatility is included as a control. Columns 2 and 3 use realized volatility over $t+1$ through $t+12$ as the outcome and construct the price path from 12 and 24 monthly returns ending in t , respectively. Log implied volatility is the log of the monthly average of daily 30-day model-free implied volatility from OptionMetrics. Realized volatility is 100 times the sample standard deviation of the twelve subsequent monthly returns. All outcomes are standardized. Panel A regresses each outcome jointly on the five individual features. Panel B reports separate regressions for the chart-risk index S^E and the sequence index S^S ; regressions with S^S also control for volatility and total return. In columns 2 and 3, the index regressions additionally control for current log implied volatility. The indices are defined in Equations (2) and (3), and the market feature definitions are reported in Internet Appendix Table IA.7. All regressions include stock and calendar-month fixed effects and control for log market capitalization at the end of the previous December. Standard errors are reported in parentheses and are two-way clustered by stock and month. For the index coefficients in Panel B, standard errors also incorporate uncertainty in the estimated experimental weights using the delta method. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

	One-month-ahead implied volatility (path to $t-1$) (1)	Future realized volatility	
		Chart window H used to build the features	
		$H=12$ (2)	$H=24$ (3)
<i>Panel A. The five features, entered jointly</i>			
Volatility (V)	0.0282*** (0.0064)	-0.0049 (0.0156)	-0.0090 (0.0215)
Recency (R)	0.0168*** (0.0044)	0.0350*** (0.0095)	0.0611*** (0.0096)
Range (G)	0.1047*** (0.0169)	0.2982*** (0.0404)	0.2958*** (0.0354)
Total return (D)	-0.1016*** (0.0121)	-0.2846*** (0.0284)	-0.2547*** (0.0266)
Autocorrelation (A)	-0.0002 (0.0015)	-0.0060 (0.0048)	0.0083 (0.0061)
<i>Panel B. Composite indices, each row is a separate regression</i>			
Chart-risk index S^E	0.0383*** (0.0068)	0.0611*** (0.0109)	0.0501*** (0.0104)
Sequence index S^S , with V and D	0.0417*** (0.0093)	0.0614*** (0.0182)	0.0719*** (0.0213)
Lagged implied volatility	Yes		
Current implied volatility, Panel B		Yes	Yes
Observations, Panel A	170,614	177,350	169,113
Observations, Panel B	162,596	166,855	159,289

retail flow.

To examine whether order-related chart content matters more when retail investors account for a larger share of trading, we also split stocks by lagged retail trade share. The cutoff is the monthly median across all stocks in the Boehmer et al. (2021) universe, computed before merging with our optionable-stock sample.²⁸

Results. The chart-risk index predicts retail net selling in all three samples. A one-standard-deviation increase in S^E predicts 0.05 standard deviations more net selling over the following year among high-share stocks, 0.07 among low-share stocks, and 0.06 in the full panel (Table 15, Panel B). The sequence index shows a different pattern: it predicts net selling only among stocks with high retail trade shares.

The feature-level results reinforce this distinction. When the five features enter jointly in the high-share sample, all carry the experimental sign and all but autocorrelation are statistically significant (Panel A, column 1). In the low-share sample, only volatility and total return significantly predict net selling with the experimental signs (Panel A, column 2).

IV.E Mutual-fund redemptions

We then examine the same demand-side question in mutual funds. Mutual-fund investors often see the net asset value (NAV) paths of the funds they hold on fund-company websites or in brokerage accounts. We ask whether subsequent redemptions are related to risk perception features of these NAV paths and, in particular, whether the order-related features predict redemptions after controlling for volatility and total return.

Data and specification. We construct the fund measures in the same way as the stock measures, replacing the stock price path with a share class's 36-month NAV path. We use 36 months because it is the shortest horizon used in Morningstar and Lipper ratings. The path is based on quoted NAVs, the prices at which investors buy and sell fund shares, and excludes distributions (Internet Appendix

²⁸Optionable stocks are larger and tend to have lower retail trade shares, so the high-share group contains 22,659 of the 101,698 stock-months (Internet Appendix D).

Table 15
Price paths and subsequent retail net selling

This table reports regressions of subsequent retail net selling on stock price-path indices and individual features for optionable stocks from 2015 to 2023. Retail net selling is sell dollars minus buy dollars over $t+1$ through $t+12$, divided by their sum and standardized. Columns 1 and 2 split observations by lagged retail trade share at the monthly median in the full Boehmer et al. (2021) universe; column 3 uses all eligible observations. Monthly retail trade share is the fraction of off-exchange trades classified as retail, and lagged share is its average over $t-12$ through $t-1$. Panel A regresses retail net selling jointly on the five individual features. Panel B reports separate regressions for the chart-risk index S^E and the sequence index S^S ; regressions with S^S also control for volatility and total return. The indices are defined in Equations (2) and (3); all indices and features use twelve monthly returns ending in t . All regressions include stock and calendar-month fixed effects and control for previous-December log market capitalization, month- t return, and trailing retail imbalance. Standard errors are reported in parentheses and are two-way clustered by stock and month. For the index coefficients in Panel B, standard errors also incorporate uncertainty in the estimated experimental weights using the delta method. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

	High lagged retail share (1)	Low lagged retail share (2)	All eligible stocks (3)
<i>Panel A. The five features, entered jointly</i>			
Volatility (V)	0.0679*** (0.0228)	0.0339** (0.0145)	0.0183 (0.0120)
Recency (R)	0.0548*** (0.0159)	-0.0094* (0.0052)	0.0040 (0.0066)
Range (G)	0.1407*** (0.0434)	0.0058 (0.0213)	0.0755*** (0.0168)
Total return (D)	-0.1496*** (0.0418)	-0.0840*** (0.0202)	-0.1216*** (0.0191)
Autocorrelation (A)	0.0031 (0.0149)	-0.0110 (0.0084)	-0.0124* (0.0072)
<i>Panel B. Composite indices, each row is a separate regression</i>			
Chart-risk index S^E	0.0482** (0.0195)	0.0672*** (0.0127)	0.0634*** (0.0114)
Sequence index S^S , with V and D	0.0624*** (0.0169)	-0.0159 (0.0125)	0.0169 (0.0113)
Observations	22,659	74,612	101,698
Stocks	814	2,028	2,258

Table IA.7).

Unlike the stock features, the fund features are standardized once rather than year by year, using all share-class months from 2000 onward for which the five features are available. One standard deviation of a fund index therefore refers to this fixed reference sample. The estimation sample includes equity-fund share classes whose funds hold at least 90 percent of their assets in equities and have at least \$1 million in total net assets. It contains 19,854 share classes of 10,082 CRSP portfolios from December 2004 through March 2025, including exchange-traded and index-fund share classes.

The main outcome is an indicator equal to one when a share class's compounded net outflow over the following twelve months exceeds 20 percent of its assets. This occurs in 24.9 percent of share-class months. We use the indicator rather than continuous flow because it focuses on large redemptions and is less sensitive to the heavy tails of fund flows (Internet Appendix Table IA.9).²⁹

Regressions include share-class and calendar-month fixed effects, control for the skewness and excess kurtosis of the NAV returns in the feature window, and use standard errors two-way clustered by CRSP portfolio and month.

Results. The chart-risk index predicts large redemptions over the following 12 months. A one-standard-deviation increase in S^E is associated with a 4.7-percentage-point higher probability of a large redemption (Table 16, Panel A, column 1). This result is robust across 12-, 24-, and 36-month windows and 10, 20, and 30 percent redemption thresholds (Panel B). The sequence index also predicts large redemptions after controlling for volatility and total return: a one-standard-deviation increase is associated with a 3.8-percentage-point higher redemption probability (Panel A, column 2). When the five features enter jointly, all features except volatility recency significantly predict large redemptions with the signs implied by the experimental estimates (Panel A, column 3).

²⁹The twelve-month compounded net flow has a standard deviation of 1.05 around a mean near zero, so continuous-flow regressions can be heavily influenced by a small number of extreme observations.

Table 16
NAV paths and large mutual-fund redemptions

This table reports linear probability regressions of large mutual-fund redemptions on NAV-path indices and individual features for CRSP equity-fund share classes. A large redemption is a compounded net outflow over $t+1$ through $t+12$ that exceeds the indicated threshold. Panel A uses a 36-month NAV history and a 20 percent threshold. Column 1 regresses the redemption indicator on the chart-risk index S^E , column 2 regresses it on the sequence index S^S together with volatility and total return, and column 3 jointly enters the five individual features. Panel B reports the S^E coefficient for 12-, 24-, and 36-month NAV histories and redemption thresholds of 10, 20, and 30 percent. NAV-path features are computed from consecutive monthly quoted NAVs ending in t ; distributions are not included in the returns used to construct the path. The indices are defined in Equations (2) and (3). All regressions include share-class and calendar-month fixed effects and control for skewness and excess kurtosis of the NAV returns in the feature window. Standard errors are reported in parentheses and are two-way clustered by CRSP portfolio and month. For the index coefficients, standard errors also incorporate uncertainty in the estimated experimental weights using the delta method. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Panel A. Main specification: 36-month window, 20 percent threshold

	Chart-risk index (1)	Sequence index with V and D (2)	Five features (3)
Volatility (V)		0.0216*** (0.0060)	0.0128*** (0.0045)
Recency (R)			0.0041 (0.0030)
Range (G)			0.0581*** (0.0063)
Total return (D)		-0.0583*** (0.0069)	-0.0803*** (0.0076)
Autocorrelation (A)			0.0069** (0.0034)
Chart-risk index S^E	0.0472*** (0.0079)		
Sequence index S^S		0.0380*** (0.0087)	

Panel B. Chart-risk index coefficients across NAV windows and redemption thresholds

NAV-history window	Outflow > 10%	Outflow > 20%	Outflow > 30%
12 months	0.0462*** (0.0071)	0.0385*** (0.0061)	0.0220*** (0.0038)
24 months	0.0500*** (0.0082)	0.0421*** (0.0072)	0.0241*** (0.0045)
36 months	0.0554*** (0.0089)	0.0472*** (0.0079)	0.0267*** (0.0047)

V Conclusion

We develop a parsimonious model of how investors perceive risk from price charts. Five features—volatility, volatility recency, price range, total return, and return autocorrelation—explain nearly as much variation in risk scores on held-out charts as the full 23-feature model and also perform well for forecast uncertainty, an incentivized measure of perceived risk. The synthetic-chart experiments further show that reordering the same returns changes perceived risk even though the realized return distribution is unchanged.

The mechanism experiments examine why the five features matter. Past price declines raise risk scores even among participants randomly assigned short positions, who benefit from further declines, suggesting that the total-return effect is not simply a response to personal losses. Participants also tend to reveal recent returns first, and the recency effect persists when reveal order is randomized, indicating that the greater weight on recent volatility does not arise merely because recent returns are viewed first. Finally, replacing the price line with return arrows largely eliminates the range effect while leaving the autocorrelation effect essentially unchanged. This contrast indicates that the two effects arise through different channels: range depends on the visual price path, whereas autocorrelation survives when price levels are removed.

We apply the experimental coefficients to market data without fitting them to any market outcome. The chart-risk index combines all five features using the experimental risk-score coefficients, while the sequence index combines the three order-related features—volatility recency, price range, and return autocorrelation—using the same weights. Both indices are associated with contemporaneous option-implied volatility and predict both next-month implied volatility and realized volatility over the following year.

Consistent with the investment choices in the experiment, higher values of both indices are associated with lower market demand: the chart-risk index predicts subsequent retail net selling and large mutual-fund redemptions, while the sequence index predicts retail net selling among stocks with high retail trade shares and fund redemptions after controlling for volatility and total return.

The market evidence does not establish that perceived risk causes these outcomes, but it shows

that a model estimated from experimental risk judgments is informative about market risk and investor demand outside the laboratory. The experiments show that risk judgments depend not only on the returns themselves, but also on when they occur and how the resulting price path is displayed. Our five-feature model provides a parsimonious way to measure these differences in perceived risk and relate them to market outcomes.

References

- Ned Augenblick, Eben Lazarus, and Michael Thaler. Overinference from weak signals and underinference from strong signals. *Quarterly Journal of Economics*, 140(1):335–401, 2025.
- Yaron Azrieli, Christopher P Chambers, and Paul J Healy. Incentives in experiments: A theoretical analysis. *Journal of Political Economy*, 126(4):1472–1503, 2018.
- Cuimin Ba, J. Aislinn Bohren, and Alex Imas. Over- and underreaction to information: Belief updating with cognitive constraints. *Quarterly Journal of Economics*, 2026. forthcoming.
- Avijit Bansal and Joshy Jacob. Impact of price path on disposition bias. *Journal of Banking & Finance*, 143:106616, 2022.
- Brad M Barber, Xing Huang, Philippe Jorion, Terrance Odean, and Christopher Schwarz. A (sub)penny for your thoughts: Tracking retail investor activity in TAQ. *The Journal of Finance*, 79(4):2403–2427, 2024.
- Ekkehart Boehmer, Charles M Jones, Xiaoyan Zhang, and Xinran Zhang. Tracking retail investor activity. *The Journal of Finance*, 76(5):2249–2305, 2021.
- J. Aislinn Bohren, Josh Hascher, Alex Imas, Michael Ungeheuer, and Martin Weber. Mental models of information: The role of attention and memory. Working paper, December 2025; previously circulated as NBER Working Paper 32149, “A Cognitive Foundation for Perceiving Uncertainty”, 2025.
- Pedro Bordalo, John J. Conlon, Nicola Gennaioli, Spencer Y. Kwon, and Andrei Shleifer. How people use statistics. *Review of Economic Studies*, 93(1):250–285, 2026.
- Charlotte Borsboom and Stefan Zeisberger. What makes an investment risky? An analysis of price path characteristics. *Journal of Economic Behavior & Organization*, 169:92–125, 2020.

- Devdeepa Bose, Henning Cordes, Sven Nolte, Judith Christiane Schneider, and Colin Farrell Camerer. Decision weights for experimental asset prices based on visual salience. *The Review of Financial Studies*, 35(11):5094–5126, 2022.
- Alex Chincó, Samuel M Hartzmark, and Abigail B Sussman. A new test of risk factor relevance. *The Journal of Finance*, 77(4):2183–2238, 2022.
- Henning Cordes, Hannes Mohrschladt, Sven Nolte, and Judith C. Schneider. The Visual Shape Score: On its predictability in the lab, the aggregated stock market, and the cross-section of stock returns. Working paper, SSRN 3927479, 2021.
- David Danz, Lise Vesterlund, and Alistair J Wilson. Belief elicitation and behavioral incentive compatibility. *American Economic Review*, 112(9):2851–2883, 2022.
- Mark Dean, Benjamin Enke, Thomas Graeber, and Pietro Ortoleva. Reference points as information. Working paper, Columbia University, Harvard University, University of Zurich, and Princeton University, 2026.
- Kresimir Demeterfi, Emanuel Derman, Michael Kamal, and Joseph Zou. More than you ever wanted to know about volatility swaps. Quantitative strategies research notes, Goldman, Sachs & Co., March 1999.
- Stephen Diacon and John Hasseldine. Framing effects and risk perception: The effect of prior performance presentation format on investment fund choice. *Journal of Economic Psychology*, 28(1):31–52, 2007.
- Darren Duxbury and Barbara Summers. On perceptions of financial volatility in price sequences. *The European Journal of Finance*, 24(7-8):521–543, 2018.
- Benjamin Enke and Thomas Graeber. Cognitive uncertainty. *Quarterly Journal of Economics*, 138(4):2021–2067, 2023.

- Benjamin Enke and Cassidy Shubatt. Quantifying lottery choice complexity. NBER Working Paper 31677, National Bureau of Economic Research, 2023.
- Tony Q. Fan, Yucheng Liang, and Cameron Peng. The inference-forecast gap in belief updating. *Econometrica*, 94(4):1279–1312, 2026.
- Markus Glaser, Zwetelina Iliewa, and Martin Weber. Thinking about prices versus thinking about returns in financial markets. *The Journal of Finance*, 74(6):2997–3039, 2019.
- Daniel Grosshans and Stefan Zeisberger. All’s well that ends well? On the importance of how returns are achieved. *Journal of Banking & Finance*, 87:397–410, 2018.
- Menglong Guan, Ryan Oprea, and Sevgi Yuksel. Beyond instrumental value: How complexity shapes information demand. Working paper, Pennsylvania State University, University of California, Berkeley, and New York University, 2025.
- Samuel M. Hartzmark, Samuel D. Hirshman, and Alex Imas. Ownership, learning, and beliefs. *Quarterly Journal of Economics*, 136(3):1665–1717, 2021.
- Felix Holzmeister, Jürgen Huber, Michael Kirchler, Florian Lindner, Utz Weitzel, and Stefan Zeisberger. What drives risk perception? A global survey with financial professionals and laypeople. *Management Science*, 66(9):3977–4002, 2020.
- Jingwen Jiang, Bryan Kelly, and Dacheng Xiu. (Re-)Imag(in)ing price trends. *The Journal of Finance*, 78(6):3193–3249, 2023.
- Zhengyang Jiang, Hongqi Liu, Cameron Peng, and Hongjun Yan. Investor memory and biased beliefs: Evidence from the field. *The Quarterly Journal of Economics*, 140(4):2749–2804, 2025.
- Ron Kaniel, Gideon Saar, and Sheridan Titman. Individual investor trading and stock returns. *The Journal of Finance*, 63(1):273–310, 2008.
- Christine Kaufmann, Martin Weber, and Emily Haisley. The role of experience sampling and graphical displays on one’s investment risk appetite. *Management Science*, 59(2):323–340, 2013.

- Mel Win Khaw, Ziang Li, and Michael Woodford. Cognitive imprecision and small-stakes risk aversion. *Review of Economic Studies*, 88(4):1979–2013, 2021.
- Harry M Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.
- Sven Nolte and Judith C Schneider. How price path characteristics shape investment behavior. *Journal of Economic Behavior & Organization*, 154:33–59, 2018.
- Ryan Oprea. Decisions under risk are decisions under complexity. *American Economic Review*, 114(12):3789–3811, 2024.
- Elise Payzan-LeNestour, Lionnel Pradier, and Tālis J Putniņš. Biased risk perceptions: Evidence from the laboratory and financial markets. *Journal of Banking & Finance*, 154:106685, 2023.
- Cassidy Shubatt and Jeffrey Yang. Tradeoffs and comparison complexity. Working paper, Harvard University and University of California, Santa Barbara; arXiv:2401.17578, first posted January 2024, revised March 2026, 2026a.
- Cassidy Shubatt and Jeffrey Yang. Valuations under tradeoff complexity. Working paper, Harvard University and University of California, Santa Barbara, 2026b.
- Matthias Unser. Lower partial moments as measures of perceived risk: An experimental study. *Journal of Economic Psychology*, 21(3):253–280, 2000.
- Elke U. Weber, Niklas Siebenmorgen, and Martin Weber. Communicating asset risk: How name recognition and the format of historic volatility information affect risk perception and investment decisions. *Risk Analysis*, 25(3):597–609, 2005.
- Stefan Zeisberger. What is risk? How investors perceive risk in return distributions. Working paper, SSRN 2811636, 2024.

INTERNET APPENDIX

Risk Perception: Model, Mechanisms, and Market Outcomes

Not for publication in the print journal

A Feature Clustering

This appendix provides additional details on the clustering procedure used in Section II.C. We group the 23 candidate features in Table 3 using agglomerative hierarchical clustering with average linkage and distance $1 - |\rho|$, where ρ is the correlation between two features across the 200 charts of the main experiment. At each step, the two clusters with the highest average absolute correlation across pairs of their members are merged. We retain merges with an average absolute correlation of at least 0.5, which leads to eleven clusters. Table IA.1 reports the full merge sequence: Panel A shows the twelve merges retained at this threshold, and Panel B shows the ten remaining merges below the threshold.

Average linkage can leave two highly correlated individual features in different clusters. Months below start, for example, has a correlation of -0.802 with total return, corresponding to a pairwise distance of 0.198. By the time these features could merge, however, total return is already part of the seven-feature cluster formed at step 11. The average absolute correlation between months below start and the seven members of that cluster is only 0.438, below the clustering threshold. The two therefore remain in separate clusters. Under single linkage, by contrast, their pairwise correlation alone would be sufficient to place them in the same cluster.

Table IA.1
Merge sequence in price-path feature clustering

This table reports the merge sequence in the hierarchical clustering of the 23 features in Table 3. Pearson correlations are calculated across the 200 charts of the main experiment. Each feature initially forms its own cluster. At each step, the two clusters with the smallest average distance merge, where distance between two features is $1 - |\rho|$ and the cluster-level distance is averaged across all feature pairs with one member from each cluster. The middle columns identify the clusters merged at each step; a composite cluster is labeled by the step in which it was formed. Mean $|\rho|$ equals one minus the merge distance, and Size is the number of features in the newly formed cluster. Panel A lists the twelve merges with mean $|\rho| \geq 0.5$; Panel B lists the ten remaining merges below this threshold.

Step	Clusters merged		Mean $ \rho $	Size
<i>Panel A. Merges retained: mean $\rho \geq 0.5$</i>				
1	Gain-to-loss ratio	Number of gains	0.965	2
2	Maximum crash	Minimum return	0.951	2
3	Maximum price	Average price	0.920	2
4	Maximum recovery	Maximum return	0.891	2
5	Total return	step 3	0.854	3
6	Minimum price	step 5	0.819	4
7	Return skewness	step 4	0.680	3
8	Price range	Mean distance from the start	0.639	2
9	Three-month trend	step 6	0.634	5
10	Return kurtosis	Jump	0.554	2
11	step 2	step 9	0.553	7
12	Run length	step 1	0.518	3
<i>Panel B. Merges below the threshold: mean $\rho < 0.5$</i>				
13	Months below start	Ever underwater	0.455	2
14	step 7	step 10	0.415	5
15	Return difference between halves	step 13	0.337	3
16	step 12	step 14	0.307	8
17	step 11	step 15	0.289	10
18	Return autocorrelation	step 8	0.286	3
19	Volatility recency	step 16	0.215	9
20	step 17	step 18	0.206	13
21	Volatility	step 19	0.175	10
22	step 20	step 21	0.172	23

B Designed Charts

The six additional experiments use the same set of sixteen designed charts: eight conditions derived from each of two base return sequences (Table IA.2). These charts replace the reshuffled charts used in the main experiment. In each additional experiment, every participant sees all eight conditions from one base sequence exactly once. Pairwise differences between conditions therefore compare responses from the same participant and complement the feature regressions for the belief measures in Table 8 and for risk scores by position in Table 9.

The eight conditions. Each base sequence contains twelve monthly returns with a mean of zero and a standard deviation of 5.06%. *Benchmark* uses the base sequence in its original order. The other seven conditions are constructed to move particular features of the resulting price path while holding other aspects of the returns fixed whenever possible.

Past and *Recent* are designed to vary volatility recency. They reorder the same twelve *Benchmark* returns so that the largest absolute returns occur mostly in the first half of the path in *Past* and in the second half in *Recent*. The two conditions therefore differ sharply in volatility recency while preserving the full return distribution and total return. Their price ranges are also nearly identical.

Peak and *Dip* are designed to generate pronounced path shapes. *Peak* arranges gains and losses so that gains are concentrated before losses, producing a rise followed by a decline; *Dip* is the sign-reversed version of *Peak*, producing the corresponding decline and recovery. Since each base sequence is symmetric around zero, both conditions are reorderings of the same *Benchmark* returns. These reorderings preserve the return distribution but substantially increase both price range and return autocorrelation relative to *Benchmark*. They therefore provide variation in these two order-related features, although they do not isolate either feature on its own.

Up and *Down* are designed to vary total return while leaving volatility unchanged. We add +1.125 and −1.125 percentage points, respectively, to every *Benchmark* return, changing the twelve-month total return from −1.4% to +12.8% and −13.9%. We then slightly reorder the shifted returns so that *Up*, *Down*, and *Benchmark* do not present participants with nearly identical

return sequences.

Finally, *Volatile* doubles every *Benchmark* return while preserving their order, thereby doubling the standard deviation of monthly returns. This transformation also widens the resulting price range.

The designed comparisons target particular features, but they do not always vary only one feature at a time. Reordering the same returns preserves the return distribution while changing their timing and the shape of the resulting price path, so it can affect several order-related features simultaneously. Price range changes in several conditions. With every chart normalized to end at 100, shifting, scaling, or reordering returns can change the distance between the highest and lowest prices. Relative to *Benchmark*, range is wider in *Up*, *Down*, *Peak*, *Dip*, and *Volatile*, but differs little between *Past* and *Recent*. We therefore report both raw pairwise differences and differences adjusted for range. The adjustment subtracts the standardized range difference between the two conditions multiplied by the estimated range coefficient from Table 8.

The two base sequences have the same mean and standard deviation but differ in the shape of their return distributions. Base sequence 1 has kurtosis 3.2 and a largest monthly move of 10%. It is the sequence used to construct the reshuffled charts in the main experiment. Base sequence 2 has kurtosis 1.9 and a largest monthly move of 8%. Using both sequences lets us assess whether the pairwise comparisons are similar across two different return distributions.

Pairwise comparisons. Table IA.3 reports differences in the two belief measures between pairs of designed conditions. For each pair, the raw difference is the response under the first condition minus the response under the second, divided by the outcome standard deviation. Positive values therefore indicate greater perceived risk under the first condition, and the differences are expressed in the same outcome units as the feature regressions. Columns 2 and 4 adjust the raw differences for price range by subtracting the standardized range difference in column 5 multiplied by the corresponding range coefficient in Table 8. No other feature is removed in this adjustment.³⁰

The *Recent* minus *Past* comparison provides the cleanest pairwise evidence on volatility recency.

³⁰Standard errors for the adjusted differences use 1,000 participant-level bootstrap resamples. Each resample re-estimates both the paired difference and the range coefficient.

Table IA.2
Designed charts in the six additional experiments

This table lists the sixteen designed synthetic charts used in the six additional experiments. Each panel contains eight conditions derived from one base return sequence; each participant views all eight conditions from one sequence and twelve real charts. Returns are listed chronologically. SD is the sample standard deviation of monthly returns, and total return is compounded over twelve months; both are in percentage points. Price range is the highest minus lowest price after normalization to a final price of 100. *Benchmark* uses the base sequence in its original order. *Past* and *Recent* reorder the same returns to concentrate the largest movements earlier or later in the path. *Peak* and *Dip* reorder the same returns to produce a pronounced peak or dip; *Dip* is the sign-reversed version of *Peak*. *Up* and *Down* add +1.125 and -1.125 percentage points, respectively, to each return and then slightly reorder the shifted sequence. *Volatile* doubles every *Benchmark* return while preserving their order.

Condition	Twelve monthly returns (%)	SD (%)	Total return (%)	Price range
<i>Panel A. Base sequence 1</i>				
<i>Benchmark</i>	-1.25, 2.5, -2.5, 1.25, -5, 10, -10, 5, -1.25, 2.5, -2.5, 1.25	5.06	-1.4	10.6
<i>Past</i>	10, -10, 5, -5, 2.5, -2.5, 1.25, -1.25, 2.5, -1.25, 1.25, -2.5	5.06	-1.4	11.6
<i>Recent</i>	2.5, -2.5, 1.25, -1.25, 2.5, -1.25, 1.25, -2.5, 5, -5, 10, -10	5.06	-1.4	11.1
<i>Peak</i>	2.5, -1.25, 2.5, 1.25, 10, 5, -10, -5, -1.25, -2.5, 1.25, -2.5	5.06	-1.4	23.1
<i>Dip</i>	-2.5, 1.25, -2.5, -1.25, -10, -5, 10, 5, 1.25, 2.5, -1.25, 2.5	5.06	-1.4	19.0
<i>Up</i>	-1.375, 2.375, -0.125, 3.625, -3.875, 6.125, -8.875, 11.125, -1.375, 2.375, -0.125, 3.625	5.06	12.8	13.9
<i>Down</i>	-3.625, 0.125, -2.375, 1.375, -11.125, 3.875, -6.125, 8.875, -3.625, 0.125, -2.375, 1.375	5.06	-13.9	20.0
<i>Volatile</i>	-2.5, 5, -5, 2.5, -10, 20, -20, 10, -2.5, 5, -5, 2.5	10.11	-5.6	22.8
<i>Panel B. Base sequence 2</i>				
<i>Benchmark</i>	-2, 3, -5, 1.62, -6, 8, -8, 6, -1.62, 5, -3, 2	5.06	-1.4	10.1
<i>Past</i>	8, -8, 6, -6, 5, -5, 2, -2, 3, -1.62, 1.62, -3	5.06	-1.4	9.5
<i>Recent</i>	3, -3, 1.62, -1.62, 5, -2, 2, -5, 6, -6, 8, -8	5.06	-1.4	8.7
<i>Peak</i>	3, -2, 5, 1.62, 8, 6, -8, -6, -1.62, -3, 2, -5	5.06	-1.4	25.0
<i>Dip</i>	-3, 2, -5, -1.62, -8, -6, 8, 6, 1.62, 3, -2, 5	5.06	-1.4	20.3
<i>Up</i>	-1.875, 3.125, -0.495, 6.125, -4.875, 7.125, -6.875, 9.125, -3.875, 2.745, -0.875, 4.125	5.06	12.8	13.0
<i>Down</i>	-4.125, 0.875, -2.745, 3.875, -9.125, 4.875, -7.125, 6.875, -6.125, 0.495, -3.125, 1.875	5.06	-13.9	18.0
<i>Volatile</i>	-4, 6, -10, 3.24, -12, 16, -16, 12, -3.24, 10, -6, 4	10.11	-5.5	21.9

Table IA.3
Paired differences between designed charts

This table reports paired differences between designed chart conditions in the Uncertainty experiment (148 participants) and the Downside experiment (150 participants). Each participant views all eight conditions from one base sequence once. Forecast uncertainty and downside probability are defined as in Table 8. For each pair, the difference is the response under the first condition minus that under the second, divided by the outcome standard deviation; positive values indicate greater perceived risk under the first condition. Columns 1 and 3 report the raw differences. Columns 2 and 4 adjust for price range by subtracting the standardized range difference in column 5 multiplied by the corresponding range coefficient in Table 8. Column 5 reports the range difference in standard-deviation units. Standard errors are reported in parentheses. The raw differences use paired *t*-test standard errors; the adjusted differences use 1,000 participant-level bootstrap resamples that re-estimate both the paired difference and the range coefficient. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Comparison	Forecast uncertainty		Downside probability		Range
	Raw (1)	Adjusted (2)	Raw (3)	Adjusted (4)	gap (SD) (5)
<i>Volatile – Benchmark</i>	0.490*** (0.070)	0.235*** (0.077)	0.231*** (0.062)	0.056 (0.059)	+2.6
<i>Recent – Past</i>	0.240*** (0.062)	0.261*** (0.060)	0.252*** (0.048)	0.267*** (0.052)	−0.2
<i>Down – Up</i>	0.191*** (0.054)	0.074 (0.055)	−0.066 (0.068)	−0.147** (0.070)	+1.2
<i>Dip – Benchmark</i>	0.304*** (0.064)	0.113* (0.061)	0.247*** (0.062)	0.115* (0.066)	+2.0
<i>Peak – Benchmark</i>	0.451*** (0.073)	0.164** (0.073)	0.296*** (0.069)	0.099* (0.058)	+3.0
<i>Dip – Peak</i>	−0.147** (0.060)	−0.051 (0.064)	−0.050 (0.067)	0.016 (0.063)	−1.0
Participants	148	148	150	150	

The two conditions reorder the same returns and differ little in price range. Forecast uncertainty is 0.240 standard deviations higher under *Recent* before the range adjustment and 0.261 higher after it; the corresponding differences in downside probability are 0.252 and 0.267. All four estimates are significant at the 1% level and have the direction implied by the positive recency coefficients in Table 8. The *Volatile* minus *Benchmark* comparison similarly raises forecast uncertainty. Its raw difference of 0.490 falls to 0.235 after adjusting for the wider price range in *Volatile*. For downside probability, the adjusted difference is 0.056 and is not statistically significant, consistent with the weak volatility coefficient for that outcome in Table 8.

The *Peak* and *Dip* comparisons with *Benchmark* vary several order-related features at once.

Both generate substantially wider price ranges and higher return autocorrelation than *Benchmark*. After removing the estimated contribution of price range, the forecast-uncertainty differences are 0.164 for *Peak* and 0.113 for *Dip*, while the downside-probability differences are 0.099 and 0.115. These adjusted differences are smaller than the raw differences, indicating that price range accounts for an important part of the response to these path shapes. The remaining differences may also reflect autocorrelation and other features of the reordered path.

The *Down* minus *Up* comparison targets total return. Although *Down* initially produces higher forecast uncertainty, the difference becomes small and statistically insignificant after adjusting for the wider price range in *Down*. Downside probability likewise provides no evidence that lower total return raises perceived risk. Thus, the pairwise comparisons reinforce the regression evidence in Table 8 that the strong total-return effect on the risk score does not carry over to the separately elicited belief measures.

Risk scores by position. The Position experiment presents the same eight designed conditions to participants randomly assigned to long or short positions. Table IA.4 reports paired differences in risk scores for the two groups, complementing the feature regressions in Panel B of Table 9. Each comparison changes risk scores in the same direction in both groups, and none of the short-minus-long differences is statistically significant.

The short-position group responds less strongly in magnitude in all four comparisons, including those unrelated to gains or losses. Its responses are 91%, 57%, 47%, and 75% as large as those of the long-position group for *Volatile* minus *Benchmark*, *Recent* minus *Past*, *Dip* minus *Peak*, and *Down* minus *Up*, respectively. The *Recent–Past* and *Dip–Peak* comparisons hold total return fixed. The smaller *Down–Up* response in the short-position group therefore appears to be part of a broader attenuation in its responses rather than a pattern specific to personal gains and losses.

Table IA.4
Paired chart differences under long and short positions

This table reports paired differences in risk scores between designed chart conditions in the Position experiment, with 105 participants in the long-position group and 93 in the short-position group. Positions are randomly assigned before participants report risk scores, which range from 0 to 10. Each participant views all eight conditions from one base sequence once. For each pair, the difference is the risk score under the first condition minus that under the second, divided by the standard deviation of risk scores within the corresponding position group. Columns 1 and 2 report the differences for the long- and short-position groups, respectively; column 3 reports short minus long. The differences are not adjusted for price range. Standard errors are reported in parentheses. Columns 1 and 2 use paired *t*-test standard errors, and column 3 uses Welch standard errors. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Dependent variable: risk score	Long position (1)	Short position (2)	Difference (3)
<i>Down – Up</i>	0.801*** (0.120)	0.600*** (0.133)	–0.200 (0.179)
<i>Volatile – Benchmark</i>	1.008*** (0.123)	0.923*** (0.126)	–0.086 (0.176)
<i>Recent – Past</i>	0.502*** (0.090)	0.288*** (0.100)	–0.214 (0.134)
<i>Dip – Peak</i>	–0.294*** (0.108)	–0.139 (0.101)	0.155 (0.148)
Participants	105	93	198

C Months Below Start

Months below start counts the months in which the price closes below its initial value. It is highly correlated with total return: the correlation is -0.80 across the 200 charts of the main experiment and -0.82 across the 116 charts used in the additional experiments. It is also the only feature in the six-feature model whose coefficient reverses sign when total return is included (Table 6), which led us to omit it in Step 3 of Section II.C. This appendix asks whether the results attributed to total return in Section III instead reflect this alternative measure of path direction. We first replace total return with months below start in the cross-measure and position regressions and then use the designed charts to distinguish the two features.

Comparisons across risk measures. Table IA.5 repeats Table 8 with months below start replacing total return. The cross-measure pattern is similar. Months below start is strongly positively associated with the risk score ($+0.128$) and more weakly with forecast uncertainty in the main experiment ($+0.022$). The latter is elicited after the risk score, so the anchoring concern discussed in Section III.A continues to apply.

When the belief measures are elicited on their own, however, months below start has little relation to either measure. Its coefficient is -0.013 for forecast uncertainty in the Uncertainty experiment ($p = 0.47$) and -0.011 for downside probability in the Downside experiment ($p = 0.31$). Thus, replacing total return with months below start reproduces the main cross-measure pattern: a strong association with risk scores but little association with the separately elicited belief measures. The substitution also reduces the within-participant R^2 for risk scores from 0.191 to 0.158.

Comparisons across positions. Table IA.6 repeats Panel B of Table 9 with months below start replacing total return. The months-below-start coefficient is positive in both the long- and short-position groups, at $+0.095$ and $+0.066$, respectively (both $p < 0.01$). The short-minus-long difference is -0.029 ($p = 0.30$), and a joint test does not reject equality of all five feature coefficients across positions ($p = 0.31$). Thus, the position result in Section III.B is unchanged when path

Table IA.5
Risk measures with months below start replacing total return

This table repeats the regressions in Table 8, replacing total return with months below start, the number of month-end prices below the initial price. Columns 1 and 2 use the main experiment (502 participants, 10,040 observations), column 3 uses the Uncertainty experiment (148 participants, 2,960 observations), and column 4 uses the Downside experiment (150 participants, 3,000 observations). The other four features are defined in Table 3. All regressions follow Equation (1) with this substitution. In the main experiment, forecast uncertainty is elicited after the risk score; in the Uncertainty and Downside experiments, the corresponding belief measure is the only risk-perception measure elicited. All variables are standardized, and all regressions include participant fixed effects. Standard errors are reported in parentheses and are two-way clustered by participant and chart. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

	Risk score	Forecast uncertainty		Downside
	(main)	(main)	(Uncertainty)	probability
	(1)	(2)	(3)	(4)
Volatility	0.041** (0.016)	0.019** (0.008)	0.040** (0.016)	0.016 (0.011)
Recency	0.079*** (0.021)	0.027*** (0.008)	0.039** (0.015)	0.042*** (0.010)
Range	0.233*** (0.027)	0.030*** (0.009)	0.099*** (0.024)	0.067*** (0.015)
Months below start	0.128*** (0.023)	0.022** (0.009)	−0.013 (0.018)	−0.011 (0.011)
Autocorrelation	0.048** (0.021)	0.026*** (0.008)	0.048** (0.023)	0.021* (0.012)
Participant fixed effects	Yes	Yes	Yes	Yes
Within-participant R^2	0.158	0.016	0.061	0.038
Observations	10,040	10,040	2,960	3,000
Participants	502	502	148	150

direction is measured by months below start rather than total return: past price weakness raises risk scores even among participants assigned to the short position.

Distinguishing the two features. The substitution results show that months below start can reproduce the main qualitative patterns associated with total return, but they do not establish which feature better captures the path-direction effect on risk scores. Two additional tests help distinguish them.

First, the months-below-start coefficient is unstable once total return is included. In the main experiment, its coefficient changes from +0.128 when total return is omitted to −0.106 when both

Table IA.6
Risk scores by position with months below start

This table repeats Panel B of Table 9, replacing total return with months below start. The Position experiment contains 105 participants in the long-position group and 93 in the short-position group. A pooled regression includes participant fixed effects, the five features, and interactions between each feature and a short-position indicator. Columns 1 and 2 report the feature coefficients for the long- and short-position groups, respectively; column 3 reports the short-minus-long differences in these coefficients. The within-participant R^2 values are calculated from separate five-feature regressions within the two groups. Months below start is the number of month-end prices below the initial price; the other four features are defined in Table 3. All nonindicator variables are standardized. Standard errors are reported in parentheses and are two-way clustered by participant and chart. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

Dependent variable: risk score	Long investors (1)	Short sellers (2)	Difference (3)
Volatility	0.064* (0.034)	0.032 (0.021)	−0.032 (0.025)
Recency	0.106*** (0.024)	0.076*** (0.024)	−0.030 (0.019)
Range	0.244*** (0.040)	0.240*** (0.034)	−0.004 (0.040)
Months below start	0.095*** (0.032)	0.066*** (0.025)	−0.029 (0.028)
Autocorrelation	0.052 (0.040)	0.045 (0.028)	−0.007 (0.032)
Participant fixed effects	Yes	Yes	Yes
Within-participant R^2	0.201	0.150	
Observations	2,100	1,860	3,960
Participants	105	93	198

features enter (Table 6), while the total-return coefficient increases in magnitude from -0.230 to -0.316 (Table 8 and Table 6). The Position experiment shows a similar pattern in the long-position group: the months-below-start coefficient falls from $+0.095$ when it replaces total return to -0.001 ($p = 0.96$) when both features enter, while the total-return coefficient is -0.132 ($p < 0.01$).

Second, the *Dip–Peak* comparison separates months below start from total return. Total return is identical across the two conditions, while *Dip* has eleven more months below start than *Peak*. If months below start had an independent positive effect on perceived risk, *Dip* should therefore appear riskier. Instead, participants in the long-position group rate *Dip* significantly less risky than *Peak*, by 0.294 standard deviations ($p < 0.01$; Table IA.4).

We next ask which five-feature specification better explains the observed *Dip–Peak* difference.

For each specification, we estimate the effects of the five features using responses to the eighteen charts other than *Dip* and *Peak*. We then use these estimates and the feature differences between *Dip* and *Peak* to predict their difference in risk scores. The gap between the observed and predicted differences is the portion that the model does not explain. In the specification that replaces total return with months below start, the months-below-start coefficient estimated from the other eighteen charts is $+0.139$. This specification predicts *Dip* to be riskier than *Peak*, opposite to the observed difference, and leaves an unexplained difference of -0.451 ($p = 0.001$). By contrast, the paper’s five-feature model leaves only -0.066 unexplained ($p = 0.55$). The total-return specification therefore accounts substantially better for the *Dip–Peak* contrast. Adding months below start also reduces the cross-validated R^2 for forecast uncertainty from 0.0132 to 0.0128 (Table 4). These results support omitting months below start from the preferred five-feature model.

D Market Data and Empirical Implementation

Data sources. The stock panel combines CRSP stock data with OptionMetrics option data. For each stock, we compute 30-day model-free implied volatility using the variance-swap replication approach of Demeterfi et al. (1999) and match the resulting measure to CRSP. Retail order flow is constructed from off-exchange TAQ trades classified at the stock-day level using the subpenny rule of Boehmer et al. (2021). The fund panel uses monthly CRSP mutual-fund data accessed through WRDS, including quoted NAVs, fund returns, total net assets, and fund identifiers; fund flows are inferred from returns and changes in total net assets.

Table IA.7 defines the market versions of the five features and the two indices, and Table IA.8 defines the outcomes and their timing. The five features follow their experimental counterparts in Table 3, with the market-specific adaptations described below. The chart-risk and sequence indices are defined in Equations (2) and (3).

The stock sample. We retain CRSP common shares listed on the NYSE, Amex, or Nasdaq (share codes 10 and 11; exchange codes 1–3). In month t , the stock price must exceed \$5 and market capitalization must exceed the 10th percentile of CRSP common-stock capitalizations in the previous December. These screens yield 10,131 stocks over 2001–2025, of which 4,025 can be matched to single-name OptionMetrics data from January 2001 through August 2025.

The regression sample further requires twelve consecutive trailing returns, implied volatility, and the previous-December size control. After dropping stocks with only one usable observation, the sample contains 3,351 stocks and 192,814 stock-months. The index regressions contain 181,638 observations because the indices begin in 2006, with 2001–2005 used for initial feature standardization.

The retail sample. We begin with optionable stock-months in the Boehmer et al. (2021) panel for which the twelve-month forward net-sell imbalance is observed. We then restrict the sample to observations with all variables required for the regressions and exclude stocks with only one

Table IA.7
Market price-path features and indices

This table defines the market implementations of the five price-path features and the chart-risk and sequence indices. Stock returns include distributions, whereas fund NAV returns exclude them. Stock features and indices are standardized using observations from 2001 through the preceding year; fund features and indices use a fixed sample of eligible share-class months from 2000 onward. The indices use the experimental risk-score coefficients and are normalized to unit standard deviation in the corresponding standardization sample.

Object	Definition
Return path	For H monthly returns $r_1, \dots, r_H$ ending in month t , let $q_0 = 1$ and $q_j = \prod_{s=1}^j (1 + r_s)$.
<i>Five features</i>	
Volatility (V)	100 times the sample standard deviation of $r_1, \dots, r_H$. Order-invariant.
Recency (R)	100 times the sample standard deviation of $r_{H/2+1}, \dots, r_H$ minus that of $r_1, \dots, r_{H/2}$. Order-dependent.
Range (G)	$\max_{0 \leq j \leq H} q_j - \min_{0 \leq j \leq H} q_j$. Order-dependent.
Total return (D)	$100(q_H - 1)$, the compounded return over the window. Order-invariant.
Autocorrelation (A)	$\sum_{j=1}^{H-1} (r_j - \bar{r})(r_{j+1} - \bar{r}) / \sum_{j=1}^H (r_j - \bar{r})^2$. Order-dependent.
<i>Indices</i>	
Chart-risk index S^E	Weighted combination of standardized V , R , G , D , and A using the experimental risk-score coefficients, as defined in Equation (2).
Sequence index S^S	Weighted combination of standardized R , G , and A using the corresponding experimental risk-score coefficients, as defined in Equation (3).

usable observation. The resulting full sample contains 2,258 stocks, of which 814 appear in the high-retail-share sample.

For stock-month t , lagged retail trade share is the average monthly fraction of off-exchange trades classified as retail over $t - 12$ through $t - 1$, requiring at least six observed months. A stock-month is classified as high-share when this measure is at or above the contemporaneous median in the full Boehmer et al. (2021) universe, calculated before matching to the optionable-stock sample. The retail data begin in January 2015, making July 2015 the first month with the required six-month history. This classification yields 22,659 high-share observations out of 101,698 regression observations.

The fund sample. We retain share classes of equity funds with at least 90 percent of holdings in equities and at least \$1 million in total net assets. The five NAV-path features are standardized using

Table IA.8
Market outcomes and timing

This table defines the market outcomes and their timing relative to the price-path feature window. Unless otherwise stated, the feature window ends in month t .

Outcome	Definition	Timing
Option-implied volatility	Log of the monthly average of daily 30-day model-free implied volatility.	Contemporaneous with the feature window ending in t . In the lagged specification, features end in $t - 1$ and implied volatility is measured in t .
Future realized volatility	100 times the standard deviation of monthly returns over $t + 1$ through $t + 12$.	Twelve months following the feature window.
Retail net-sell imbalance	Retail sell dollars minus buy dollars over $t + 1$ through $t + 12$, divided by their sum over the same period.	Twelve months following the feature window.
Fund large redemption	Indicator that the twelve-month forward net outflow defined in Equation (IA.1) exceeds the specified threshold; the main specification uses 20 percent.	Twelve months following the NAV feature window.

all eligible share-class months from 2000 onward. For share class c , month t 's inferred flow is

$$f_{c,t} = \frac{\text{TNA}_{c,t} - \text{TNA}_{c,t-1}(1 + r_{c,t}^{\text{TR}})}{\text{TNA}_{c,t-1}},$$

where TNA is total net assets and r^{TR} is the CRSP total return, including distributions. In contrast, the NAV-path features are constructed from quoted NAVs and therefore exclude distributions. Compounded twelve-month forward net flow is

$$F_{c,t}^{12} = \prod_{k=1}^{12} (1 + f_{c,t+k}) - 1. \quad (\text{IA.1})$$

and requires twelve consecutive monthly flow observations. Monthly inferred flows below -100% or above $1,000\%$ are excluded as extreme values.

The estimation sample runs from December 2004, the first month with more than one qualifying share class, through March 2025, the last month with twelve subsequent flows available. After applying these filters and dropping fixed-effect singletons, the sample contains 1,640,738 share-class

Table IA.9
Summary statistics for the market samples

This table reports summary statistics for the stock, retail, and fund samples. N is the number of observations, SD is the sample standard deviation, and P10, P50, and P90 are percentiles. Price-path features and indices are defined in Table IA.7, and market outcomes are defined in Table IA.8. The five features are reported in their market units: V , R , and D in percentage points, G in cumulative gross-return units, and A as a unitless measure.

	N	Mean	SD	P10	P50	P90
<i>Panel A. Stock sample</i>						
Volatility (V)	192,814	11.4362	7.4911	5.2572	9.7287	19.2175
Recency (R)	192,814	0.0645	8.7234	-7.0288	0.0702	7.2984
Range (G)	192,814	0.5982	0.6856	0.2203	0.4478	1.0442
Total return (D)	192,814	22.1230	66.7669	-33.0702	12.8267	78.4731
Autocorrelation (A)	192,814	-0.1085	0.2383	-0.4182	-0.1113	0.2036
Chart-risk index S^E	181,638	-0.0707	0.9250	-1.0232	-0.1433	1.0823
Sequence index S^S	181,638	-0.0629	1.0585	-1.0106	-0.1916	0.8872
Log implied volatility	192,814	3.7359	0.4531	3.1750	3.7016	4.3509
Future realized volatility (12m)	177,350	10.7327	5.7660	5.1954	9.4253	17.7770
<i>Panel B. Retail sample</i>						
Volatility (V)	101,698	11.2810	7.5737	5.3121	9.6426	18.7951
Recency (R)	101,698	0.0792	9.2175	-6.9634	0.1316	7.2966
Range (G)	101,698	0.5844	0.7396	0.2138	0.4296	1.0123
Total return (D)	101,698	20.9492	68.2687	-32.3227	10.7442	76.5117
Autocorrelation (A)	101,698	-0.1233	0.2358	-0.4290	-0.1262	0.1860
Chart-risk index S^E	101,698	-0.0242	0.9521	-0.9859	-0.0876	1.1434
Sequence index S^S	101,698	-0.0453	1.1557	-1.0117	-0.1929	0.9171
Forward 12m net-sell imbalance	101,698	0.0088	0.0359	-0.0340	0.0077	0.0519
<i>Panel C. Fund sample</i>						
Volatility (V)	1,640,738	5.6377	6.6145	3.3261	5.2408	7.6500
Recency (R)	1,640,738	0.1058	9.3396	-2.7359	0.0370	3.1548
Range (G)	1,640,738	0.5013	0.4620	0.2637	0.4511	0.7457
Total return (D)	1,640,738	12.6672	46.5143	-24.1815	10.2703	47.9484
Autocorrelation (A)	1,640,738	-0.0581	0.1845	-0.2765	-0.0772	0.2038
Chart-risk index S^E	1,640,738	0.0317	0.9859	-0.9234	-0.1271	1.3127
Sequence index S^S	1,640,738	0.0165	0.9899	-0.7620	-0.1289	1.0072
Large redemption indicator (20%)	1,640,738	0.2488	0.4323	0.0000	0.0000	1.0000

months.

Table IA.9 reports summary statistics for the stock, retail, and fund samples.

Experimental coefficients and market implementation. In the experiment, price range is measured after the final price is normalized to 100. Applying the same normalization to rolling

Table IA.10
Experimental weights for the market indices

This table reports the experimental regressions used to estimate the coefficients entering the market indices. The sample contains 10,040 responses from 502 participants in the main experiment. The dependent variable is the standardized risk score in column 1 and standardized forecast uncertainty in column 2; all five features are also standardized. The features follow the market definitions in Table IA.7. Column 1 provides the five coefficients used for S^E and the R , G , and A coefficients used for S^S . Superscript $\dagger$ identifies these three sequence features. All regressions include participant fixed effects. Standard errors are reported in parentheses and are two-way clustered by participant and chart. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

	Risk score (1)	Forecast uncertainty (2)
Volatility (V)	0.0403*** (0.0133)	0.0184** (0.0079)
Recency (R) [†]	0.0457** (0.0177)	0.0232*** (0.0081)
Range (G) [†]	0.1401*** (0.0206)	0.0196** (0.0086)
Total return (D)	−0.3044*** (0.0242)	−0.0414*** (0.0084)
Autocorrelation (A) [†]	0.0821*** (0.0186)	0.0310*** (0.0080)
Within R^2	0.1882	0.0172
Observations	10,040	10,040
Participants	502	502
Charts	200	200

market windows can produce extreme values when a stock ends the window near zero. We instead normalize the cumulative gross-return path to one at the start of each market window and measure the difference between its highest and lowest values. The two range measures correlate between 0.25 and 0.38 across market windows and at 0.892 on the experimental charts. The other four features are equivalent to their experimental counterparts up to scale, which is removed by standardization.

Table IA.10 reports the regressions used to estimate the experimental coefficients entering the market indices. These regressions use the market definition of price range. Feature z -scores are computed across the 10,040 participant-chart observations, so charts receive weight in proportion to the number of responses they contribute rather than equal weight across the 200 charts. The resulting specification explains experimental risk scores nearly as well as the original feature definitions, with within- R^2 of 0.188 versus 0.191.

The coefficients entering S^E and S^S are estimated rather than known. We account for this estimation uncertainty in the standard errors of the market regressions using the delta method. For S^E , the calculation uses the covariance matrix of the five experimental risk-score coefficients; for S^S , it uses the corresponding covariance matrix for R , G , and A . Experimental and market sampling errors are treated as independent.

Retail trade classification. We classify off-exchange retail trades using the subpenny rule of Boehmer et al. (2021), which signs trades just above a round penny as retail sells and those just below as retail buys. Barber et al. (2024) document substantial classification error in this rule and show that a quote-based alternative improves directional accuracy. That alternative requires trade-level TAQ data that are not available for our panel, so we adopt the original classification.

E Additional Market Results

Experimental and option-market coefficients. The experimental risk-score coefficients in Table IA.10, column 1, and the corresponding option-implied-volatility coefficients in Table 13, column 3, have a cosine similarity of 0.905. For each test year, we also estimate the option-market coefficients using only earlier data. Volatility, recency, range, and total return have the same signs as their experimental counterparts in all 20 years, while the autocorrelation coefficient has the experimental sign in 14 years and is close to zero otherwise.

Model fit under experimental weights. Table IA.11 asks how much model fit is lost when the experimental weights are used instead of estimating five separate feature coefficients from the market outcomes. A specification that keeps volatility and total return separate and combines recency, range, and autocorrelation into S^S achieves 88.1% of the unrestricted five-feature model's within- R^2 for option-implied volatility and 69.4% for future realized volatility. Using S^E alone achieves about 55% for both outcomes. Thus, the experimental weights capture a substantial share of the fit of the unrestricted market regressions.

Robustness. Table IA.12 examines 6-, 12-, and 24-month price-path windows. Panel A reports the S^E and S^S coefficients for option-implied and future realized volatility, and Panel B reports the five features jointly for option-implied volatility. The results are highly robust across horizons, with stable coefficient signs and nearly all estimates remaining statistically significant.

We also examine whether the results depend on using risk-score coefficients to construct S^E . We instead construct S^C using the five forecast-uncertainty coefficients in column 2 of Table IA.10 and repeat the corresponding stock and retail regressions in Table IA.13. The correlation between the two indices is 0.89, and the coefficients on S^C remain positive and statistically significant in those regressions. This similarity shows that the stock and retail findings are not specific to using risk-score coefficients to construct the index.

Table IA.11
Model fit under experimental weight restrictions

This table compares model fit for log option-implied volatility and future realized volatility across five specifications. V and D denote volatility and total return. S^E combines all five features using the experimental risk-score coefficients; the V, D, S^S specification keeps volatility and total return separate and combines recency, range, and autocorrelation using their experimental coefficients. The five-feature specification enters all five features separately and serves as the benchmark. Relative fit is each specification's within- R^2 as a percentage of the five-feature model's within- R^2 . The features and indices are defined in Table IA.7. For this model-fit comparison, all five specifications for a given outcome use the same estimation sample. The five features are standardized using the means and standard deviations from that sample, and S^E and S^S are scaled using the same observations. This differs from the expanding historical standardization used in the main regressions in Tables 13 and 14. All regressions include stock and calendar-month fixed effects and control for previous-December log market capitalization.

Specification	Log implied volatility		Future realized volatility	
	Within R^2	Relative fit (%)	Within R^2	Relative fit (%)
V	0.1665	61.7	0.0207	36.7
V, D	0.1974	73.2	0.0277	49.0
S^E	0.1485	55.1	0.0310	54.9
V, D, S^S	0.2375	88.1	0.0392	69.4
Five features entered separately	0.2697	100.0	0.0565	100.0
Observations	192,814		177,350	

Table IA.12
Market regressions across price-path windows

This table reports stock regressions using price-path windows of 6, 12, and 24 months ending in t . Panel A reports separate regressions for the chart-risk index S^E and the sequence index S^S using standardized log implied volatility and future realized volatility as outcomes; regressions with S^S also include standardized volatility and total return. The future-realized-volatility regressions additionally control for current log implied volatility, as in Table 14. Panel B reports the five-feature specification for standardized log implied volatility. The features and indices are defined in Table IA.7, and the outcomes are defined in Table IA.8. All regressions include stock and calendar-month fixed effects and control for previous-December log market capitalization. Standard errors are reported in parentheses and are two-way clustered by stock and month. For the index coefficients in Panel A, standard errors also incorporate uncertainty in the estimated experimental weights using the delta method. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

	$H = 6$	$H = 12$	$H = 24$
<i>Panel A. Composite indices, each row is a separate regression</i>			
Log option-implied volatility			
Chart-risk index S^E	0.1437*** (0.0232)	0.1515*** (0.0208)	0.1331*** (0.0181)
Sequence index S^S , with V and D	0.0888*** (0.0178)	0.1389*** (0.0269)	0.1818*** (0.0385)
Observations	186,416	181,638	172,534
Future realized volatility			
Chart-risk index S^E	0.0476*** (0.0117)	0.0611*** (0.0109)	0.0501*** (0.0104)
Sequence index S^S , with V and D	0.0471*** (0.0120)	0.0614*** (0.0182)	0.0719*** (0.0213)
Observations	170,560	166,855	159,289
<i>Panel B. Five features entered jointly, log option-implied volatility</i>			
Volatility (V)	0.1066*** (0.0177)	0.1433*** (0.0170)	0.2198*** (0.0257)
Recency (R)	0.0166*** (0.0057)	0.0707*** (0.0154)	0.1092*** (0.0144)
Range (G)	0.2079*** (0.0417)	0.2813*** (0.0388)	0.2330*** (0.0289)
Total return (D)	-0.2479*** (0.0282)	-0.2946*** (0.0250)	-0.2352*** (0.0216)
Autocorrelation (A)	0.0023 (0.0028)	0.0077*** (0.0028)	0.0234*** (0.0036)
Observations	197,800	192,814	182,884

Table IA.13
Market regressions with forecast-uncertainty weights (S^C)

This table reports market regressions using S^C , which weights the five features by the forecast-uncertainty coefficients in column 2 of Table IA.10. S^C is constructed analogously to S^E : for an observation in year y , the features are standardized using observations from 2001 through $y - 1$, and the resulting index is scaled to unit standard deviation in the same sample. The dependent variable is log option-implied volatility in column 1, future realized volatility in columns 2 and 3, and retail net selling in columns 4 and 5. Column 3 additionally controls for current implied volatility, while columns 4 and 5 use the high-retail-share and full retail samples, respectively. All outcomes are standardized. The features and outcomes are defined in Tables IA.7 and IA.8. All regressions include stock and calendar-month fixed effects and control for previous-December log market capitalization. The retail regressions additionally control for month- t return and trailing retail imbalance. Standard errors are reported in parentheses and are two-way clustered by stock and month, with uncertainty in the experimental coefficients incorporated using the delta method. *, **, and *** denote significance at the 10 percent, 5 percent, and 1 percent levels, respectively.

	Option-implied volatility	Future realized volatility		Retail net selling	
	Full sample (1)	Baseline (2)	+ IV (3)	High share (4)	Full sample (5)
S^C	0.1452*** (0.0323)	0.0852*** (0.0171)	0.0389*** (0.0106)	0.0505* (0.0276)	0.0399*** (0.0126)
Observations	181,638	166,855	166,855	22,659	101,698